eLife

Reviewed Preprint
v1 • August 13, 2024
Not revised

Cell Biology

Oxidized low-density lipoprotein potentiates angiotensin II-induced Gq activation through the AT1-LOX1 receptor complex: Implications for renal dysfunction

Jittoku Ihara, Yibin Huang, Yoichi Takami, Yoichi Nozato, Toshimasa Takahashi, Akemi Kakino, Cheng Wang, Ziwei Wang, Yu Guo, Weidong Liu, Nanxiang Yin, Ryoichi Ohara, Taku Fujimoto, Shino Yoshida, Kazuhiro Hongyo, Hiroshi Koriyama, Hiroshi Akasaka, Hikari Takeshita, Shinsuke Sakai, Kazunori Inoue, Yoshitaka Isaka, Hiromi Rakugi, Tatsuya Sawamura, Koichi Yamamoto

Department of Geriatric and General Medicine, Osaka University Graduate School of Medicine, 2-2 Yamada-oka, Suita, Osaka, 565-0871, Japan • Center for Pulmonary and Vascular Biology, Dept. of Pediatrics, University of Texas Southwestern Medical Center, 5323 Harry Hines Blvd., Dallas, TX, 75390, USA • Department of Medicine, Takahashi clinic, Kobe, Hyogo, 657-0846, Japan • Department of Molecular Pathophysiology, Shinshu University Graduate School of Medicine, Matsumoto, Nagano 390-8621, Japan • Department of Nephrology, Osaka University Graduate School of Medicine, 2-2 Yamada-oka, Suita, Osaka, 565-0871, Japan

Abstract

Chronic kidney disease (CKD) and atherosclerotic heart disease, frequently associated with dyslipidemia and hypertension, represent significant health concerns. We investigated the interplay among these conditions, focusing on the role of oxidized low-density lipoprotein (oxLDL) and angiotensin II (Ang II) in renal injury via G protein αq subunit (Gq) signaling. We hypothesized that oxLDL enhances Ang II-induced Gq signaling via the AT1 (Ang II type 1 receptor)-LOX1 (lectin-like oxLDL receptor) complex. Based on CHO and renal cell model experiments, oxLDL alone did not activate Gq signaling. However, when combined with Ang II, it significantly potentiated Gq-mediated inositol phosphate 1 production and calcium influx in cells expressing both LOX-1 and AT1 but not in AT1-expressing cells. This suggests a critical synergistic interaction between oxLDL and Ang II in the AT1-LOX1 complex. Conformational studies using AT1 biosensors have indicated a unique receptor conformational change due to the oxLDL-Ang II combination. In vivo, wild-type mice fed a high-fat diet with Ang II infusion presented exacerbated renal dysfunction, whereas LOX-1 knockout mice did not, underscoring the pathophysiological relevance of the AT1-LOX1 interaction in renal damage. These findings highlight a novel mechanism of renal dysfunction in CKD driven by dyslipidemia and hypertension and suggest the therapeutic potential of AT1-LOX1 receptor complex in patients with these comorbidities.

eLife assessment

This study provides **useful** in vitro evidence to support a mechanism whereby dyslipidemia could accelerate renal functional decline through the activation of the AT1R/LOX1 complex by oxLDL and AngII. As such, it improves the knowledge regarding the complex interplay between dyslipidemia and renal disease and provides a **solid** basis for the discovery of novel therapeutic strategies for patients with lipid disorders. The methods, data, and analyses support the presented findings, although the observed variability and need for further in vivo validation require additional research in this key area.

https://doi.org/10.7554/eLife.98766.1.sa2

Introduction

Dyslipidemia is a major risk factor of atherosclerotic heart disease in patients with chronic kidney disease (CKD). It has been postulated that the association between dyslipidemia and CKD is not solely a result of epidemiological comorbidities but rather a complex interplay of causality, where CKD exacerbates dyslipidemia, while dyslipidemia, in turn, contributes to the onset and progression of CKD[1,2]. Since Moorhead et al. proposed the lipid nephrotoxicity hypothesis in 1982[3], accumulating evidence has suggested that increased plasma lipid levels contribute to the development of renal glomerular and tubular damage, primarily in animal models of dyslipidemia[4]. The etiology of dyslipidemia-induced nephrotoxicity is complex and multifaceted, potentially involving the activation of certain cellular signaling pathways that culminate in renal injury through elevated levels of oxidized LDL (oxLDL)[5–8]. The lectin-like oxLDL receptor, LOX-1, is implicated in organ damage caused by dyslipidemia, and its expression is increased in hypertensive glomerulosclerosis[9]. In murine models, a deficiency of LOX-1 leads to a reduction in renal dysfunction, which was precipitated by a systemic inflammatory state following significant myocardial ischemia and injury[10]. This supports the hypothesis that LOX-1 plays a role in the development of inflammation-induced renal injury. However, to date, no studies have investigated the role of LOX-1 in nephrotoxicity caused by dyslipidemia.

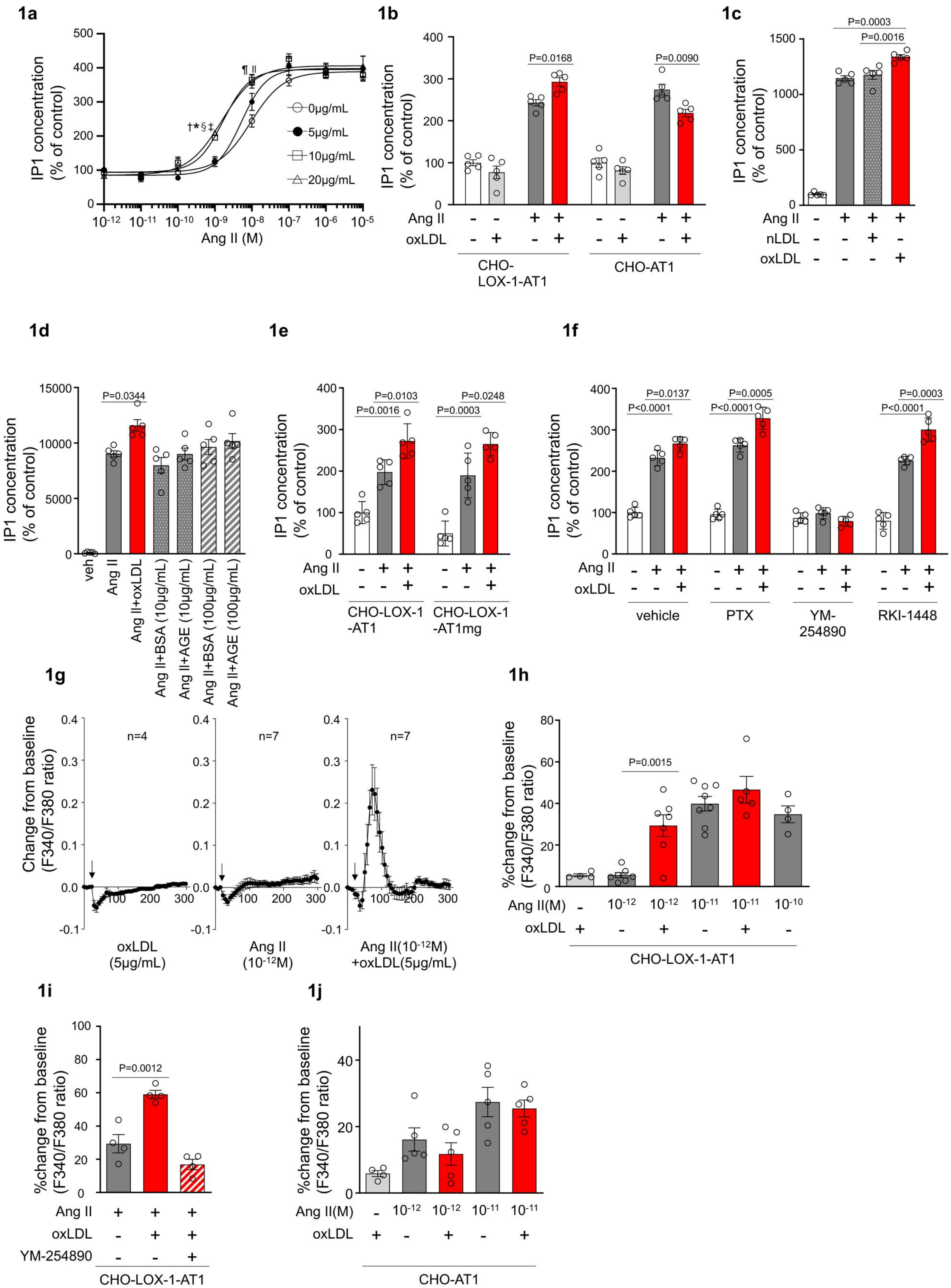
Hypertension is an established risk factor for CKD, and it has been suggested that hypertension and dyslipidemia act synergistically to induce renal dysfunction[11]. The development of renal damage due to hypertension involves direct renal injury by vasoactive hormones, such as angiotensin II (Ang II), in addition to renal hemodynamic abnormalities associated with elevated body pressure[12]. We have shown that LOX-1 and the Ang II type 1 receptor (AT1) of the G protein-coupled receptor (GPCR) are coupled on the plasma membrane, and that G protein-dependent and β-arrestin-dependent AT1 activation mechanisms are involved in the signaling mechanism by oxLDL and the intracellular uptake of oxLDL, respectively[13,14].

Interestingly, it was recently demonstrated that AT1 exhibits different modes and degrees of G protein activation depending on the conformational changes that occur during activation by various ligands[15]. Specifically, G protein αq subunit (Gq)-biased agonists induce a more open conformation of AT1 than Ang II, resulting in a more potent activation of Gq, which is the primary mediator of Ang II-induced hypertension. In contrast, β-arrestin-biased agonists induce a closer conformational change, leading to the activation of β-arrestin without the activation of Gq[15]. Indeed, we found that oxLDL selectively activates G protein αi subunit (Gi) of AT1, without activating Gq[14], similar to that induced by β-arrestin-biased agonists[16]. However, given that Ang II and oxLDL coexist in physiological environments, it is plausible to hypothesize that the effect of these ligands on AT1 in living organisms may differ from their effects when administered individually. Indeed, we found that the combination treatment of oxLDL and Ang II enhanced pro-inflammatory NFκB activity compared to each treatment alone in cells overexpressing both AT1 and LOX-1[14]. Based on our findings and the aforementioned structure-activity relationship of AT1, we hypothesized that the binding of both oxLDL and Ang II to LOX-1 and AT1, respectively, may result in a more open AT1 structure, leading to stronger downstream Gq signaling. Therefore, this study aimed to investigate and clarify this hypothesis, focusing specifically on renal component cells, and ultimately demonstrate the in vivo relevance of this phenomenon in the development of renal injury or renal dysfinction under conditions of Ang II and oxLDL overload.

Results

Oxidized LDL potentiates Ang II-induced Gq signaling in a LOX-1-dependent manner

First, we examined the additive effect of oxLDL on Ang II-stimulated AT-1-Gq signaling in CHO (Chinese hamster ovary) cells that did not endogenously express LOX-1 and AT-1 but were genetically engineered to express these receptors (CHO-LOX-1-AT1)[13]. **Fig. 1a** shows the dose-response curve of Ang II-induced IP1 production, which serves as a measure of Gq activity, in the presence of varying concentrations of oxLDL in CHO-LOX-1-AT1 cells. Consistent with previous findings, oxLDL alone did not stimulate IP1 production in CHO-LOX-1-AT1 cells[14]. However, when oxLDL was supplemented at concentrations of 10 and 20 μg/mL (but not at 5 μg/mL), it caused a similar leftward shift in the dose-response curve of Ang II, resulting in a more than 80% decrease in the Effective Concentration 50 (EC50) (EC50 values: 0 μg/mL, 9.40×10^{-9} M; 5 μg/mL, 5.21×10^{-9} M; 10 μg/mL, 1.68×10^{-9} M; 20 μg/mL, 1.55×10^{-9} M). The maximum IP1 production induced by Ang II in CHO-LOX-1-AT1 cells was unaffected by the addition of oxLDL.



1a
IP1 concentration (% of control)
Ang II (M)
0μg/mL
5μg/mL
10μg/mL
20μg/mL
†*§‡
¶ ||
1b
IP1 concentration (% of control)
P=0.0168
P=0.0090
Ang II
oxLDL
CHO-LOX-1-AT1
CHO-AT1
1c
P=0.0003
P=0.0016
Ang II
nLDL
oxLDL
1d
P=0.0344
veh
Ang II
Ang II+oxLDL
Ang II+BSA (10μg/mL)
Ang II+AGE (10μg/mL)
Ang II+BSA (100μg/mL)
Ang II+AGE (100μg/mL)
1e
P=0.0103
P=0.0016
P=0.0248
P=0.0003
CHO-LOX-1-AT1
CHO-LOX-1-AT1mg
1f
P=0.0137
P<0.0001
P=0.0005
P<0.0001
P=0.0003
P<0.0001
vehicle
PTX
YM-254890
RKI-1448
1g
Change from baseline (F340/F380 ratio)
n=4
n=7
n=7
oxLDL (5μg/mL)
Ang II (10-12M)
Ang II(10-12M) +oxLDL(5μg/mL)
1h
%change from baseline (F340/F380 ratio)
P=0.0015
Ang II(M)
oxLDL
CHO-LOX-1-AT1
1i
P=0.0012
Ang II
oxLDL
YM-254890
CHO-LOX-1-AT1
1j
Ang II(M)
oxLDL
CHO-AT1

Fig. 1

Oxidized LDL potentiates Ang II-induced Gq signaling and calcium influx in a LOX-1-dependent manner

a Dose-dependent response of IP1 concentration by the activation of Gq signaling in response to oxLDL and Ang II in CHO-LOX-1-AT1 cells. Cells were treated with oxLDL and Ang II at the concentrations described in the figure. (n=4 for each oxLDL concentration). **P*=0.0004 for 0 μg/mL vs 20 μg/mL, †*P*=0.0003 for 5 g/mL vs 20 μg/mL, ‡P=0.0020 for 0 μg/mL vs 10 μg/mL, and§*P*=0.0015 for 5 μg/mL vs 10 μg/mL at 10^{-9} M Ang II; ||*P*=0.0051 for 0 μg/mL vs 10 μg/mL, ¶*P*=0.0004 for 0 μg/mL vs 20 μg/mL at 10^{-8} M Ang II.

b IP1 concentration in response to vehicle, native LDL (nLDL 10 μg/mL), and oxLDL (10 μg/mL) in the combination of Ang II (10^{-8} M) in CHO-LOX-1-AT1 cells (n=5 for each group)

c IP1 concentration in response to vehicle, oxLDL (10 μg/mL) in the combination of Ang II (10^{-8}M) in CHO-LOX-1-AT1 and CHO-AT1 cells (n=5 for each group).

d IP1 concentration in response to vehicle, oxLDL (10 μg/mL), BSA (10 or 100 μg/mL), BSA-conjugated AGE (10 or 100 μg/mL) in the combination of Ang ll (10^{-8} M) in CHO-LOX-1-AT1 cells.

e IP1 concentration in response to vehicle, oxLDL (10 μg/mL) in the combination of Ang II (10^{-8} M) in genetically engineered CHO cells with or without intact β-arrestin binding domain (n=5 for each group). AT1mg indicates AT1 a mutant AT1 lacking a functional β-arrestin binding domain but retaining G-protein-biased signaling capability.

f IP1 concentration in response to oxLDL (10 μg/mL) in the combination of Ang II (10^{-8} M) and additional effect of PTX, a Gi inhibitor, YM-254890, a Gq inhibitor, and RKI-1448, a downstream Rho kinase inhibitor targeting G12/13 signaling, in CHO-LOX-1-AT1 cells (n=5 for each group).

g Intracellular calcium dynamics measured using Fura 2-AM by the ratio of the mission signals at excitation wavelength 340 nm and 380 nm in response to oxLDL (5 μg/mL), Ang II (10^{-12} M), and their combination (for each agonist, 4-7 regions of interest were selected). Addition of these agonists is marked with arrows on the timeline of the assay.

h Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with oxLDL and Ang II at specified concentrations in CHO-LOX-1-AT1 cells as detailed in the figure (n=4-8).

i Percentage change from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM after stimulation with oxLDL (5 μg/mL), Ang II (10^{-12} M), and YM-254890, a Gq inhibitor, in CHO-LOX-1-AT1 cells.

j Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with oxLDL (5 μg/mL) and Ang II at specified concentrations in CHO-AT1 cells, as detailed in the figure.

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a-f) and (h-j).

Additionally, we observed that oxLDL administration decreased Ang II-induced IP1 production in CHO cells expressing AT1 alone (CHO-AT1), in contrast to CHO cells expressing both AT1 and LOX-1 (**Fig. 1b**). This finding suggests that the potentiating effect of oxLDL on Ang II-AT1-Gq activity is dependent on the presence of LOX-1. However, the reason for the decrease in IP1 production caused by oxLDL supplementation was not determined in this study. Native LDL, which does not bind to LOX-1, did not alter Ang II-induced Gq activity in CHO-LOX-1-AT1 cells (**Fig. 1c**). Furthermore, the presence of advanced glycation end products (AGEs) that bind to LOX-1[17], did not enhance Ang II-induced IP1 production (**Fig. 1d**).

We found that enhanced IP1 production by co-treatment of oxLDL with Ang II was similarly observed in CHO-cells expressing LOX-1 and mutated AT1 with impaired ability to activate β arrestin[14] (**Fig 1e**). Moreover, the potentiating effect of oxLDL on IP1 production was unaffected by Pertussis Toxin (PTX), a Gi inhibitor, or RKI-1448, a downstream Rho kinase inhibitor targeting G12/13 signaling in CHO-LOX-1-AT1 cells. However, this phenomenon was completely inhibited by YM-254890, the Gq inhibitor (**Fig. 1f**). These results suggest that the potentiating effect of oxLDL on Ang II-induced IP1 production is not influenced by β-arrestin, Gi, or G12/13, which are the main effectors of AT1 signaling aside from Gq[18].

oxLDL potentiates Ang II-induced calcium influx in a LOX-1-dependent manner

Calcium influx is a representative cellular phenomenon that occurs in response to Ang II-AT1-Gq activation. We found that oxLDL and low concentrations of Ang II (10^{-12} M) did not induce calcium influx in CHO-LOX-1-AT1 cells when treated alone, but apparently induced calcium influx when treated together (**Fig. 1g, h**). Ang II alone at higher concentrations (10^{-11} M) induced calcium influx, and no further enhancement was observed with oxLDL supplementation (**Fig. 1h**). The combined effect of oxLDL and Ang II on calcium influx was completely blocked by YM254890, suggesting that this phenomenon was Gq-dependent (**Fig. 1i**). Importantly, oxLDL supplementation with Ang II did not affect the calcium influx in CHO cells expressing AT1 alone (**Fig. 1j**).

Co-treatment of oxLDL with Ang II induces conformational change of AT1 different from each treatment alone

To gain mechanistic insight into this phenomenon, we initially conducted a live-imaging analysis of membrane LOX-1 and AT1 to determine whether the combined treatment of oxLDL and Ang II leads to increased internalization of AT1 upon activation compared to individual treatments, as described in a previous protocol paper[19] (**Fig. S1**, Supplemental Video, and **Fig. 2a**).

Our findings revealed a decrease in green puncta, which represent AT1-eGFP, upon treatment with Ang II, oxLDL alone, and their co-treatment compared with the control group (**Fig. 2a**). However, the extent of membrane AT1 reduction was similar across all the treatment groups (**Fig. 2a**). Consistent with our previous report[14], oxLDL treatment resulted in a reduction in the red puncta representing LOX-1-mScarlet (**Fig. 2a**). Importantly, co-treatment with oxLDL and Ang II did not further enhance the reduction of red puncta compared to oxLDL treatment alone (**Fig. 2a**). Based on these findings, it is conceivable that co-treatment with oxLDL and Ang II does not increase the content of activated AT1 compared to individual treatments alone.

AT1 intramolecular FlAsH-BRET biosensors were used to detect AT1 conformational changes in CHO cells expressing LOX-1[20]. Among several biosensors previously tested [20], we used two sensors with FlAsH insertion at the third intracellular loop (ICL3P3) and cytoplasmic-terminal tail (C-tailP1) of AT1 that interacts with Renilla luciferase (RlucII) at the end of the cytoplasmic tail (AT1-ICL3P3 and AT1-C-tailP1, respectively), as these sensors were shown to enhance BRET signaling induced by Ang II compared to biased agonists, including SI, SII, DVG, and SBpA[20]. In CHO cells expressing LOX-1 alone (CHO-LOX-1) transduced with lentivirus encoding AT1-C-tailP1, 10^{-5} M Ang II and the combination of Ang II and 10 μg/mL oxLDL induced BRET similarly, whereas oxLDL alone did not alter BRET (**Fig. 2b**). In contrast, in CHO-LOX-1 cells transduced with AT1-ICL3P3-encoded lentivirus, the combination of Ang II and oxLDL induced BRET more prominently than Ang II alone, whereas oxLDL alone did not alter BRET (**Fig. 2c**). Furthermore, the

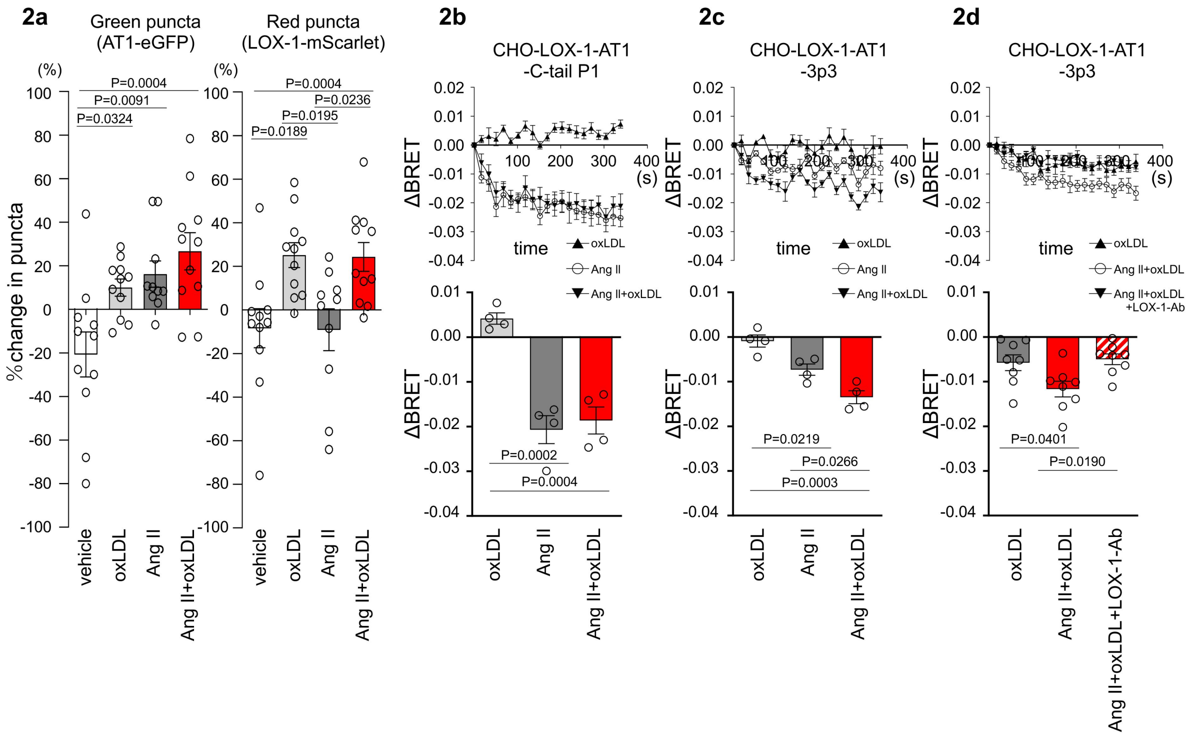


Fig. 2

Co-treatment of oxLDL with Ang II induces conformational change of AT1 different from each treatment alone

a Change in green puncta (AT1-eGFP) and red puncta (LOX-1-mScarlet) by the treatment with vehicle, oxLDL (10μg/mL), Ang II (10^{-5}M), and the combination of oxLDL and Ang II in CHO cell overexpressing these fluorescent protein-conjugated receptors (n=10-11 for each group). The puncta were manually counted by a blinded observer and the number of puncta at 0 and 3 min was determined (N0 and N3, respectively). The change in puncta was calculated as (N0-N3)/N0.

b-d Changes in BRET signals were monitored in CHO-LOX-1 cells expressing the following conformational biosensors: AT1-C-tailP1 (b) and AT1-ICL3P3 (c, d) bearing FlAsH insertion at the cytoplasmic-terminal tail (C-tailP1) and the third intracellular loop (ICL3P3) of AT1, respectively, that interact with Renilla luciferase at the end of the cytoplasmic tail. Cells were subjected to treatments with vehicle, oxLDL (10 μg/mL), Ang II (10^{-5}M), the combination of Ang II and oxLDL, and the combination of AngII, oxLDL and LOX-1 antibody. The BRET ratios were calculated every 16 seconds for a total of 320 seconds and the relative change in intramolecular BRET ratio (ΔBRET) was calculated by subtracting the average BRET ratio measured for cells stimulated with vehicle at each time point. Lower panels indicate average ΔBRET of all the time points during measurement. Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-d).

difference between oxLDL and the combination treatment was abolished by a neutralizing antibody against LOX-1 (**Fig. 2d**). These findings suggested that the concomitant binding of oxLDL to LOX-1 and Ang II to AT1 induced a conformational change in AT1 that was distinct from that induced by Ang II or oxLDL alone.

Oxidized LDL potentiates Ang II-induced Gq-calcium signaling in renal cells

To confirm the pathophysiological significance of this phenomenon observed in the overexpressing cells, we validated it in cells endogenously expressing LOX-1 and AT1. We found that oxLDL in combination with Ang II did not increase the cellular IP1 content in human umbilical vein endothelial cells (HUVECs), bovine vascular endothelial cells (BACEs), human aortic vascular smooth muscle cells (HAVSMCs), or rat macrophages (A10) (**Fig. S2**). In contrast, in normal rat kidney epithelial cells (NRK52E) and fibroblasts (NRK49F), the combination of oxLDL and Ang II, but not Ang II alone, increased IP1 accumulation, which was suppressed in the presence of YM-25480, the Gq inhibitor (**Fig. 3a, b**). IP1 accumulation induced by co-treatment with Ang II and oxLDL was abolished by the siRNA-mediated knockdown of either AT1 or LOX-1 (**Fig. 3 c, d**; the knockdown efficacy is shown in **Fig. S3**). Regarding calcium influx, intracellular calcium concentrations were not increased by the treatment of either 10^{-7} M Ang II or low concentration of oxLDL (2 μg/ml) alone (**Fig. 3e**, left and middle, **Fig 3f**). In contrast, the combination of Ang II (10^{-7} M) and oxLDL (2 μg/ml) increased intracellular Ca^{2+} concentration (**Fig. 3e**, right, **Fig. 3f**). The combined effect on calcium influx was attenuated by the siRNA knockdown of either AT1 or LOX-1 (**Fig. 3g**). The Gq inhibitor and angiotensin receptor blocker (ARB), Irbesartan, also inhibited this phenomenon (**Fig. 3h**). Calcium influx was not induced by either combination therapy or monotherapy in NRK52E cells (**Fig. S4**).

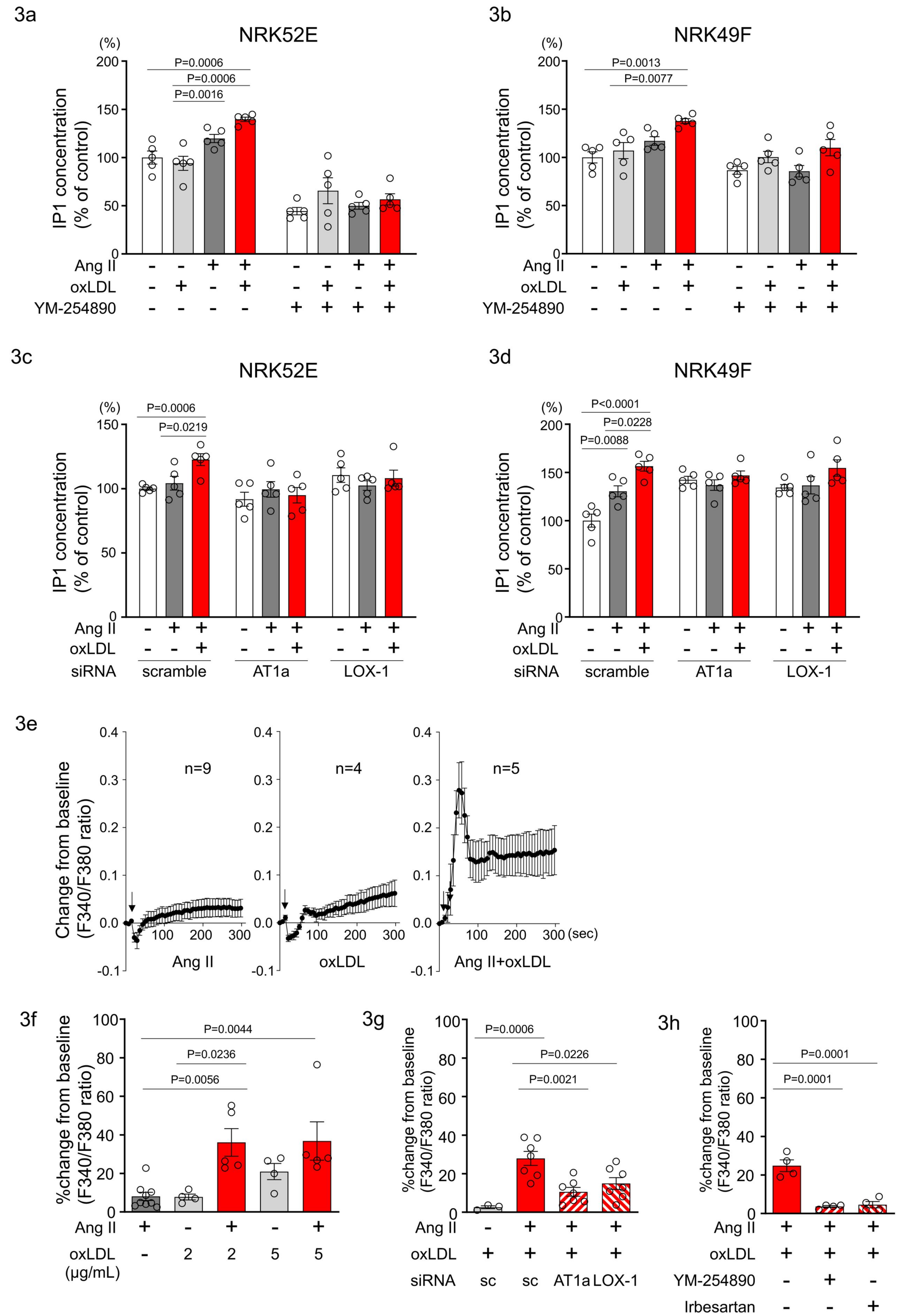

3a
NRK52E
IP1 concentration (% of control)
P=0.0006
P=0.0006
P=0.0016
Ang II
oxLDL
YM-254890
3b
NRK49F
P=0.0013
P=0.0077
3c
NRK52E
P=0.0006
P=0.0219
siRNA
scramble
AT1a
LOX-1
3d
NRK49F
P<0.0001
P=0.0228
P=0.0088
3e
Change from baseline (F340/F380 ratio)
n=9
n=4
n=5
Ang II
oxLDL
Ang II+oxLDL
(sec)
3f
%change from baseline (F340/F380 ratio)
P=0.0044
P=0.0236
P=0.0056
oxLDL (μg/mL)
3g
P=0.0006
P=0.0226
P=0.0021
sc
3h
P=0.0001
P=0.0001
Irbesartan

Fig. 3

Oxidized LDL potentiates Ang II-induced Gq-calcium signaling in renal cells

a, b IP1 concentration in response to Ang II (10^{-7} M), oxLDL (10 μg/mL), and the combination of both with or without YM-254890, Gq inhibitor, in NRK52E (a) and NRK49F cells (b) (n=5 for each group).

c, d IP1 concentration in response to Ang II (10^{-7} M) and oxLDL (10 μg/mL) in the combination of Ang II (10^{-7} M) and the additional effect of siRNA-mediated knockdown of *AT1a* or *LOX-1* in NRK52E (c) and NRK49F cells (d) (n=5 for each group).

e Intracellular calcium concentration in NRK49F cells using Fura 2-AM and dual-excitation microfluorometry. Changes in the fluorescence intensity ratio (F340/F380) served as an index of the calcium dynamics. Cells were exposed to Ang II (10^{-7} M), oxLDL (2 μg/mL), and a combination of both agents. Addition of these agonists is marked with arrows on the timeline of the assay. Data acquisition and analysis were performed using a digital image analyzer to monitor real-time calcium flux (n=4-9).

f Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL at the concentrations detailed in the figure in NRK49F cells (n=5 for each group).

g Impact of siRNA-mediated knockdown *AT1* or *LOX-1* on the percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL (2 μg/mL) in NRK49F cells (n=3-7).

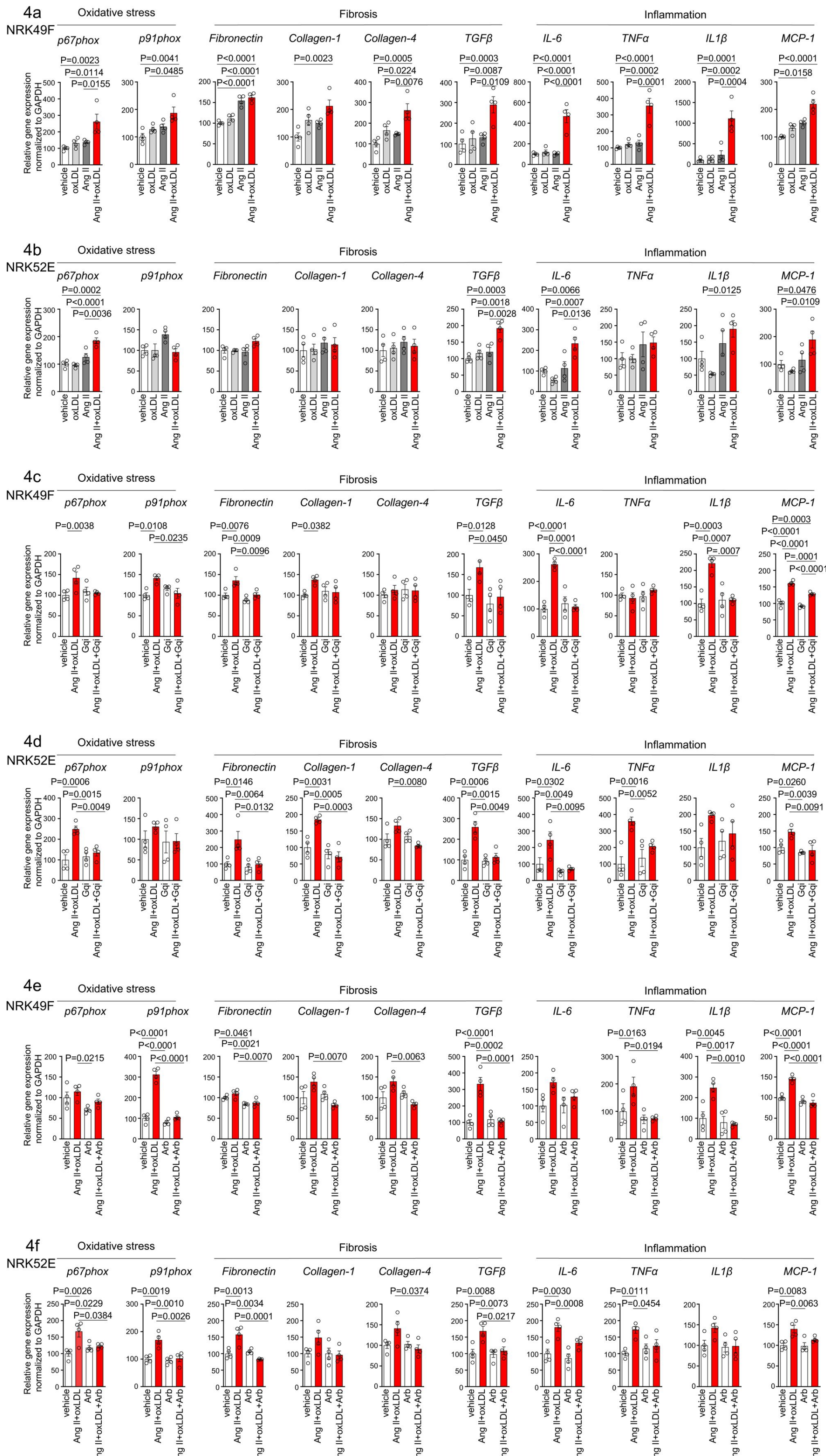
g Impact of co-treatment of YM-254890, Gq inhibitor, or Irbesartan, an angiotensin receptor blocker (ARB), on the the percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7} M) and oxLDL (2 μg/mL) in NRK49F cells (n=4 for each group).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a-d) and (f-h).

Co-treatment of oxLDL and Ang II enhanced cellular response upon Gq activation in renal cells

In NRK49F cells, co-treatment of oxLDL and Ang II, compared to vehicle, increased mRNA levels of NADPH components, *p67phox* and *p91phox*, fibrosis markers, *fibronectin*, *collagen-1*, *collagen-4*, and *TGFβ*, and inflammatory cytokines, *TNFα*, *IL1β*, *IL-6,* and *MCP-1* (**Fig. 4a**). Notably, oxLDL did not alter the expression of the genes of interest, and Ang II increased the mRNA levels of only *fibronectin* and *MCP-1*, indicating the apparent synergistic effect of co-treatment with Ang II and oxLDL on specific gene expression. This amplification effect of co-treatment was also observed in limited genes including *p67phox*, *TGFβ*, *IL-6,* and *MCP-1* in NRK52E cells (**Fig. 4b**). The combined effects of oxLDL and Ang II on gene expression were completely abolished by the Gq inhibitor (**Fig. 4c, d**) and ARB (**Fig. 4e, f**) in NRK49F and NRK 52E. We also verified protein expression of αSMA as a molecular marker of epithelial-mesenchymal transition (EMT) upon the indicated stimulation for 3 days in rat kidney cells. The combination of Ang II (10^{-7} M) and oxLDL (5 μg/mL) induced αSMA expression in NRK49F or NRK52E to the same extent or less than TGFβ, a major inducer of EMT, respectively (**Fig. 5a, b**). The results of the combination treatment were strikingly different from that of oxLDL or Ang II treatment alone, which did not affect αSMA expression in both types of cells (**Fig. 5c, d**). The induction of αSMA by the combined treatment was suppressed in the presence of a Gq inhibitor or ARB in both types of cells, suggesting the AT1-

Gq-dependent pathway in this phenomenon (**Fig. 5e, f** and **Fig. 5g, h**). As a final in vitro assay, the proliferative activity of NRK49F cells after 24 h of treatment with Ang II and oxLDL, either alone or in combination, was measured using the BrdU assay (**Fig. 6a**). When administered alone, Ang II and oxLDL increased and reduced BrdU incorporation, respectively. The reducing effect of oxLDL on proliferation was blocked by Irbesartan, but not by a Gq inhibitor, consistent with our findings that oxLDL induces biased activation of AT1, which favors Gi but not Gq signaling[14]. Oxidized LDL potentiated the Ang II-induced increase in BrdU incorporation, which was blocked in the presence of a Gq inhibitor or an ARB (**Fig. 6a**). The siRNA-mediated knockdown of AT1 completely abolished the effects of Ang II and oxLDL, either alone or in combination. Knockdown of LOX-1 did not affect the pro-proliferative effect of Ang II itself but blocked the enhanced proliferation induced by co-treatment with oxLDL (**Fig. 6b**).



4a NRK49F
Oxidative stress
Fibrosis
Inflammation
p67phox
p91phox
Fibronectin
Collagen-1
Collagen-4
TGFβ
IL-6
TNFα
IL1β
MCP-1
Relative gene expression normalized to GAPDH
vehicle
oxLDL
Ang II
Ang II+oxLDL
4b NRK52E
4c NRK49F
Gqi
Ang II+oxLDL+Gqi
4d NRK52E
4e NRK49F
Arb
Ang II+oxLDL+Arb
4f NRK52E

Fig. 4

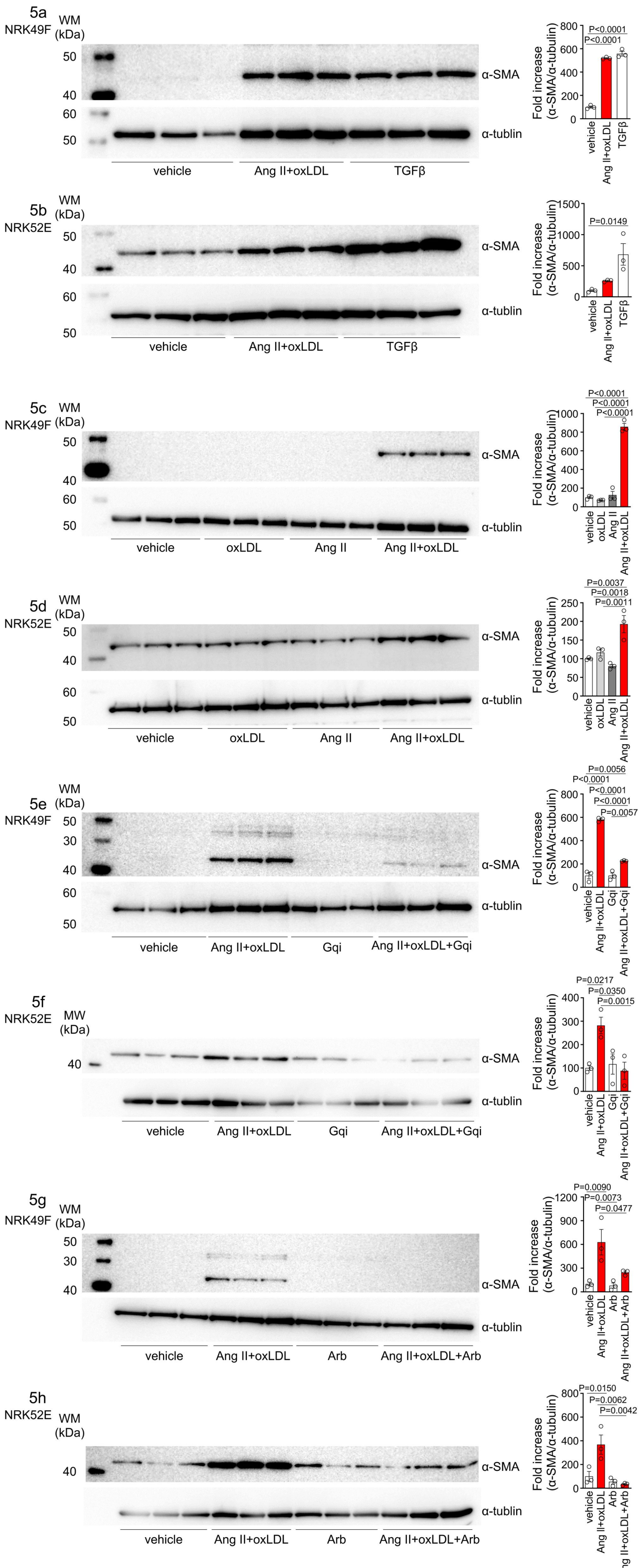
Co-treatment of oxLDL and AII enhanced cellular response upon Gq activation in renal cells

a,b Quantitative real-time PCR analysis was performed to measure the gene expression of NADPH oxidase subunits (*p67phox* and *p91phox*), fibrosis markers (*fibronectin*, *collagen-1*, *collagen-4*, and *TGFβ*), and inflammatory cytokines (*TNFα*, *IL1β*, *IL-6*, and *MCP-1*) in NRK49F cells (a) and NRK 52E cells (b). Gene expression levels were normalized to those of GAPDH. Cells were stimulated by oxLDL (5 μg/mL), Ang II (10^{-7} M) or their combination (n=4 for each group).

c, d Cells were pre-treated with vehicle or Gq inhibitor (YM-254890, Gqi), followed by treatment with vehicle or the combination of oxLDL (5 μg/mL) and Ang II (10^{-7} M) in NRK49F cells (c) and NRK 52E cells (d) (n=4 for each group).

e, f Cells were pre-treated with vehicle or ARB (Irbesartan, Arb), followed by treatment with vehicle or the combination of oxLDL (5 μg/mL) and Ang II (10^{-7} M) in NRK49F cells (e) and NRK 52E cells (f) (n=4 for each group).

Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-f).



5a
NRK49F
WM (kDa)
50
40
60
50
α-SMA
α-tublin
vehicle
Ang II+oxLDL
TGFβ
Fold increase (α-SMA/α-tubulin)
P<0.0001
P<0.0001
5b
NRK52E
WM (kDa)
α-SMA
α-tublin
vehicle
Ang II+oxLDL
TGFβ
P=0.0149
5c
NRK49F
WM (kDa)
α-SMA
α-tublin
vehicle
oxLDL
Ang II
Ang II+oxLDL
P<0.0001
P<0.0001
P<0.0001
5d
NRK52E
WM (kDa)
α-SMA
α-tublin
vehicle
oxLDL
Ang II
Ang II+oxLDL
P=0.0037
P=0.0018
P=0.0011
5e
NRK49F
WM (kDa)
50
30
40
α-SMA
α-tublin
vehicle
Ang II+oxLDL
Gqi
Ang II+oxLDL+Gqi
P=0.0056
P<0.0001
P<0.0001
P<0.0001
P=0.0057
5f
NRK52E
MW (kDa)
40
α-SMA
α-tublin
vehicle
Ang II+oxLDL
Gqi
Ang II+oxLDL+Gqi
P=0.0217
P=0.0350
P=0.0015
5g
NRK49F
WM (kDa)
α-SMA
α-tublin
vehicle
Ang II+oxLDL
Arb
Ang II+oxLDL+Arb
P=0.0090
P=0.0073
P=0.0477
5h
NRK52E
WM (kDa)
40
α-SMA
α-tublin
vehicle
Ang II+oxLDL
Arb
Ang II+oxLDL+Arb
P=0.0150
P=0.0062
P=0.0042

Fig. 5

Oxidized LDL enhanced Ang II-induced epithelial mesenchymal transition in NRK52E and NRK49F cells

a, b Left: Western blot analysis of α-smooth muscle actin (α-SMA), a marker of epithelial–mesenchymal transition (EMT), in NRK49F (a) and NRK52E (b) cells. Cells were stimulated with oxLDL (5 μg/mL), Ang II (10^{-7} M), and TGF-β (10 ng/mL), with TGF-β serving as a well-known EMT inducer. Right: Densitometric analysis of α-SMA protein expression normalized to α-Tubulin (n=3 for each group).

c, d Left: Western blot analysis of α-SMA in NRK49F (c) or NRK52E (d) after stimulation with oxLDL (5 μg/mL), Ang II (10^{-7} M), and their combination. Right: Densitometric analysis of α-SMA protein expression normalized to α-tubulin (n=3 for each group).

e, f Left: Western blot analysis of α-SMA in NRK49F(e) or NRK52E (f) after treatment with a combination of oxLDL (5 μg/mL) and Ang II (10^{-7} M). Prior to this treatment, the cells were pre-treated with either a vehicle or a Gq inhibitor (YM-254890, Gqi). Right: Densitometric analysis of α-SMA protein expression normalized to α-Tubulin (n=3 for each group).

g, h Left: Western blot analysis of α-SMA in NRK49F (e) or NRK52E (f) after treatment with a combination of oxLDL (5 μg/mL) and Ang II (10^{-7} M). Prior to treatment, cells were pre-treated with either vehicle or ARB (Irbesartan, Arb). Right: Densitometric analysis of α-SMA protein expression normalized to α-tubulin (n=3 for each group).

Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey's multiple comparison test (a-f).

Oxidized LDL-inducible diet exacerbates Ang II-induced renal dysfuntion in wildtype mice, but not in LOX-1 knockout mice

To examine the relevance of this phenomenon in renal injury, we replicated the in vivo conditions that occurred during simultaneous stimulation with oxLDL and Ang II in wildtype (WT) and LOX-1 knockout (LOX-1 KO) mice. This was achieved by inducing oxLDL via a high-fat diet (HFD) [21] and Ang II via a subcutaneously implanted osmotic mini-pump (**Fig. 7a**). Two different doses of Ang II were used in the experiment: a pressor dose of 0.7 γ, which was demonstrated to increase blood pressure (BP) and induce renal dysfunction[22],[23], and a subpressor dose of 0.1 γ, which does not affect BP. For the intended analysis, the results are presented separately for mice exposed to subpressor and pressor doses of Ang II, utilizing the same outcomes of control animals infused with saline for comparison. Consistent with a previous report, the HFD used in the study prominently increased the plasma LOX-1 ligand concentration, which was undetectable under a normal diet (ND) after 6 weeks of feeding (**Fig. S5**). Notably, the HFD did not intensify but rather attenuated the body weight increase during the experimental period compared to the ND (**Fig. S6a-c**). The pressor dose of Ang II similarly increased BP in HFD-fed and ND-fed WT mice (**Fig. 7b**, **Fig. S6d**). However, notably, there was a modest trend of attenuated BP elevation by 0.7 γ Ang II infusion in LOX-1 KO mice, in line with the previous report showing reduced Ang II-induced BP elevation by LOX-1 deficiency in mice (**Fig. 7b**, **Fig. S6d, f**)[24]. As expected, a subpressor dose of Ang II did not alter BP in the corresponding mouse group (**Fig. 7b**, **Fig. S6e**). Regarding biofluid analysis, HFD with saline infusion did not alter the urinary 8-OHdG concentration as a marker of oxidative stress and urinary albumin excretion (UAE) compared to ND with saline infusion in either WT or LOX-1 KO mice (**Fig. 7c, d**). The pressor dose of Ang II with ND significantly increased the urinary 8-OHdG concentration and UAE in WT mice (**Fig. 7c, d**). Notably, HFD feeding in WT mice with a pressor dose of Ang II resulted in a prominent

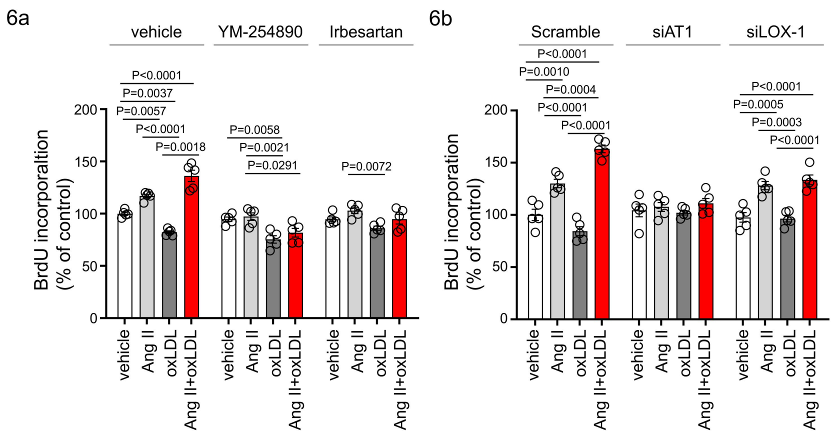


Fig. 6

Oxidized LDL enhanced Ang II-induced renal fibroblast proliferation via AT1-Gq signaling and LOX-1-dependent manner

a Proliferative activity assessed by BrdU incorporation into NRK49F cells. Cells were pretreated with vehicle, YM-254890, or ARB, Irbesartan, followed by the treatment with oxLDL (5 μg/mL), Ang II (10^{-7} M), or their combination. (n=5 for each group).
b NRK49F cells were subjected to siRNA-mediated knockdown using specific siRNAs for *AT1a* (siAT1) or *LOX-1* (siLOX-1). Following knockdown, cells were treated with either vehicle, oxLDL (5 μg/mL), Ang II (10^{-7} M), or their combination. Proliferative activity was assessed by measuring the BrdU levels (n=5 for each group).
Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

increase in urinary 8-OHdG concentration and UAE compared to mice fed with ND (**Fig. 7c, d**). In WT mice administered with a subpressor dose of Ang II, a significant increase in UAE was observed when comparing the effects of HFD to ND (**Fig. 7d**). There were no significant differences in urinary 8-OHdG levels between the two dietary conditions (**Fig. 7c**). Interestingly, when LOX-1 KO mice were fed either HFD or ND and then administered the corresponding dose of Ang II, no differences were observed in the measured parameters (**Fig. 7c, d**). These findings indicate that the combination of HFD and Ang II administration appears to have a more pronounced effect on certain biofluid markers of renal injury in WT mice than in LOX-1 KO mice. The presence or absence of LOX-1 appears to influence the interaction between HFD and Ang II, affecting these specific parameters in mice. We did not find a difference in plasma aldosterone concentration between ND- and HFD-fed WT mice with a pressor dose of Ang II (**Fig. S7**).

We then conducted quantitative real-time PCR analysis on genes within kidney sections, encompassing NADPH components (*p40phox, p47phox, p67phox, and p91phox*), inflammatory cytokines (*IL-6, TNFα, IL1β, MCP-1, and COX-2*), fibrosis markers (*TGFβ, fibronectin, collagen-1a, collagen-4a, αSMA, and vimentin*), epithelial markers (*E-cadherin and cadherin-16*), and tubular marker (*NAGL*) (**Fig. 8**, **Fig S8**). In mice treated with a pressor dose of Ang II, 15 out of 18 genes examined showed enhanced alterations in gene expression, indicative of heightened NADPH oxidase components, inflammatory cytokines, fibrosis, decreased epithelial markers, and exacerbated tubular injury in HFD-fed WT mice compared to ND-fed mice (**Fig. 8a**, **Fig. S8a**). For many of these genes, a synergistic effect of the combination of a pressor dose of Ang II and HFD was not observed in LOX-1 KO mice (**Fig. 8a**, **Fig. S8a**). Seven genes displayed discernible differences between HFD- and ND-fed WT mice treated with a subpressor dose of Ang II (**Fig. 8b** **Fig. S8b**). Other than *fibronectin*, no differences were observed between ND- and HFD-fed LOX-1 mice exposed to a subpressor dose of Ang II (**Fig. 8b** **Fig. S8b**). In relation to AT1 and LOX-1 expression, no variations emerged between the ND and HFD groups when subjected to the corresponding dose of Ang II treatment in both WT and LOX-1 KO mice (**Fig. S8a, b**).

In the histological analysis of kidney samples, in contrast to the gene expression data, the administration of either a subpressor or pressor dose of Ang II, both in isolation and in combination with HFD for 4 weeks, did not reveal any notable increase in fibrosis, as evaluated by Masson-Trichrome staining (**Fig. S9a**). Similarly, there was no significant change in the degree of mesangial expansion or glomerular area, as assessed by PAS staining. (**Fig. S9b**).

Finally, immunofluorescence staining of mouse kidney specimens was performed to detect the colocalization sites of LOX-1 and AT1 in the kidney. As shown in **Fig. 9a** and **Fig. 9b**, LOX-1 and AT1a were predominantly colocalized in the renal tubules, but not in the glomeruli.

Discussion

We conducted a series of experiments to provide empirical support for our hypotheses. We propose that the simultaneous binding of Ang II to AT1 and oxLDL to LOX-1 triggers distinct and more pronounced structural modifications in AT1 than the individual modifications induced by each ligand. This structural alteration in AT1 leads to the enhanced activation of Gq signaling.

Our experiments showed that oxLDL enhanced the effects of Ang II-induced production of IP1, a downstream signaling molecule associated with Gq activation and subsequent calcium influx. However, this effect was observed only in the presence of both LOX-1 and AT1. Considering that oxLDL selectively activates Gi but not Gq through the LOX-1-AT1 dependent pathway[14], it is evident that the observed phenomenon cannot be attributed solely to the additive effect of oxLDL on AT1 activation. Rather, the simultaneous binding of oxLDL and Ang II to their respective receptors, LOX-1 and AT1, which form a single complex, underlies this phenomenon. Indeed, our findings from live imaging of the membrane receptors revealed that the combination of Ang II and

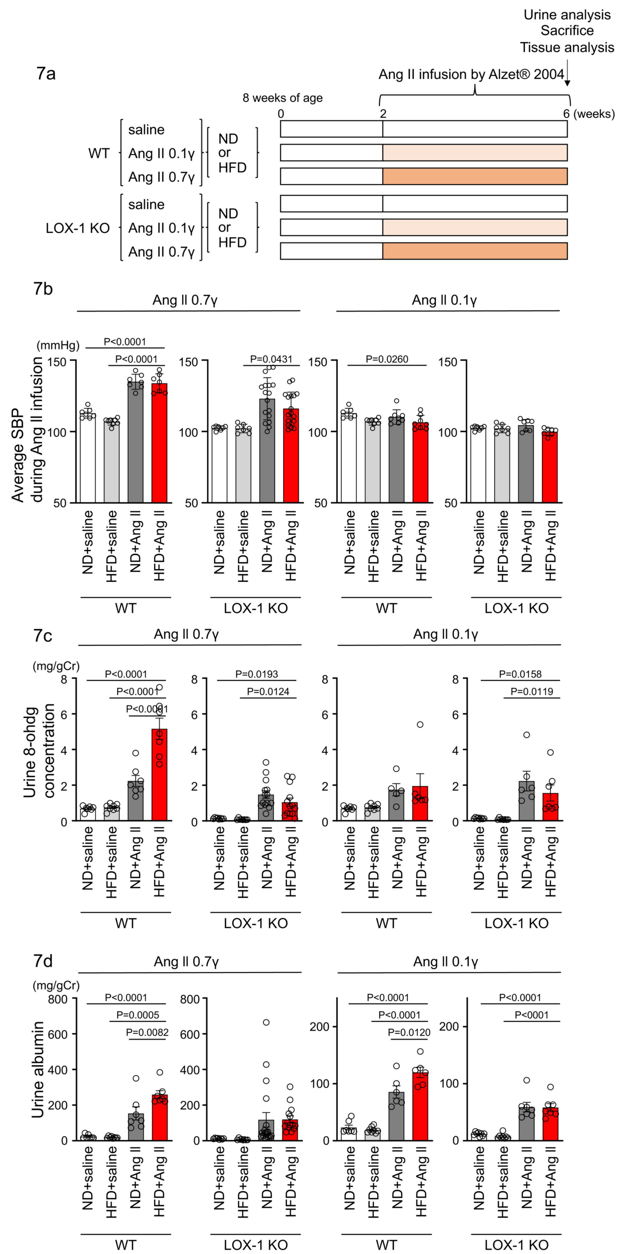


Fig. 7

Oxidized LDL-inducible diet exacerbates Ang II-induced renal dysfuntion in wildtype mice, but not in LOX-1 knockout mice

a A Schematic protocol for the animal experiments. Eight-week-old male WT mice and male LOX-1 KO mice were fed either an ND or an HFD for 6 weeks. After 10 weeks of age, the mice were treated over a 4-week period with infusions of either saline or Ang II. Ang II was administered at two dosage levels: a subpressor dose of 0.1 γ and a pressor dose of 0.7 γ, delivered via subcutaneously implanted osmotic pumps. At the end of the infusion period, urine was collected, the animals were sacrificed, and comprehensive tissue analysis was conducted to evaluate the renal effects of the treatments.

b Average systolic blood pressure (SBP) measured at half-week intervals in WT and LOX-1 KO mice during the 4-week infusion period.

c, d Urine 8-OHDG concentrations (mg/g creatinine [Cr]) (c) and urine albumin concentrations (mg/g creatinine [Cr]) (d) in WT and LOX-1 KO mice at the conclusion of the 4-week infusion period.

Data are represented as mean ± SEM. Differences were determined by one-way ANOVA, followed by Tukey’s multiple comparison test (a-d).

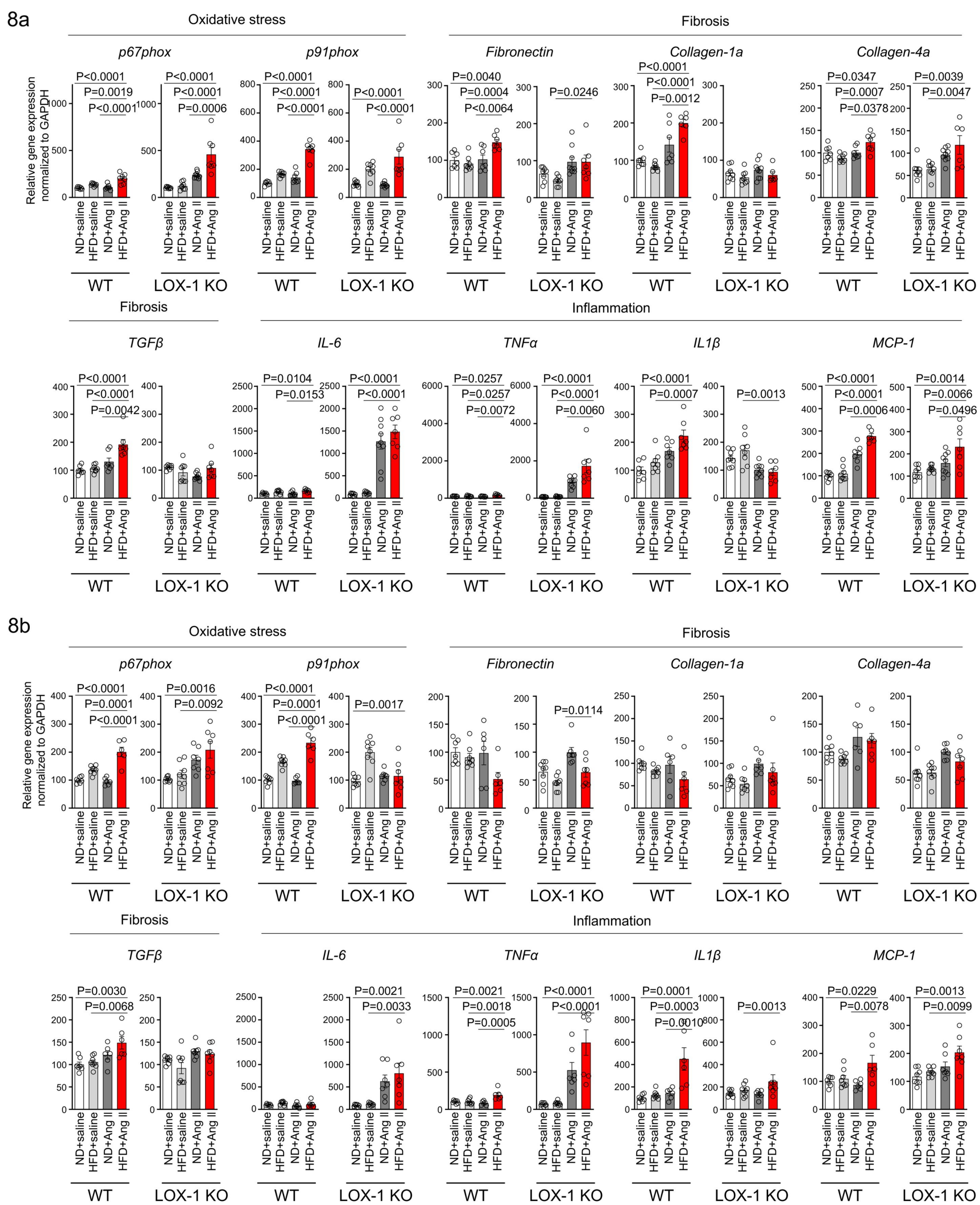


Fig. 8

A high-fat diet enhanced Ang II-induced renal injury-related gene expression in the kidney in a LOX-1-dependent manner.

Quantitative real-time PCR analysis for gene expression of NADPH components (*p67phox and p91phox*), inflammatory cytokines (*IL-6, TNFα, IL1β, and MCP-1*), and fibrosis markers (*TGFβ, fibronectin, collagen-1a, and collagen-4a*) in the kidney harvested from WT and LOX-1 KO mice.

The experimental procedures, including the dietary regimen and Ang II administration, are detailed in **Fig. 7a**.

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

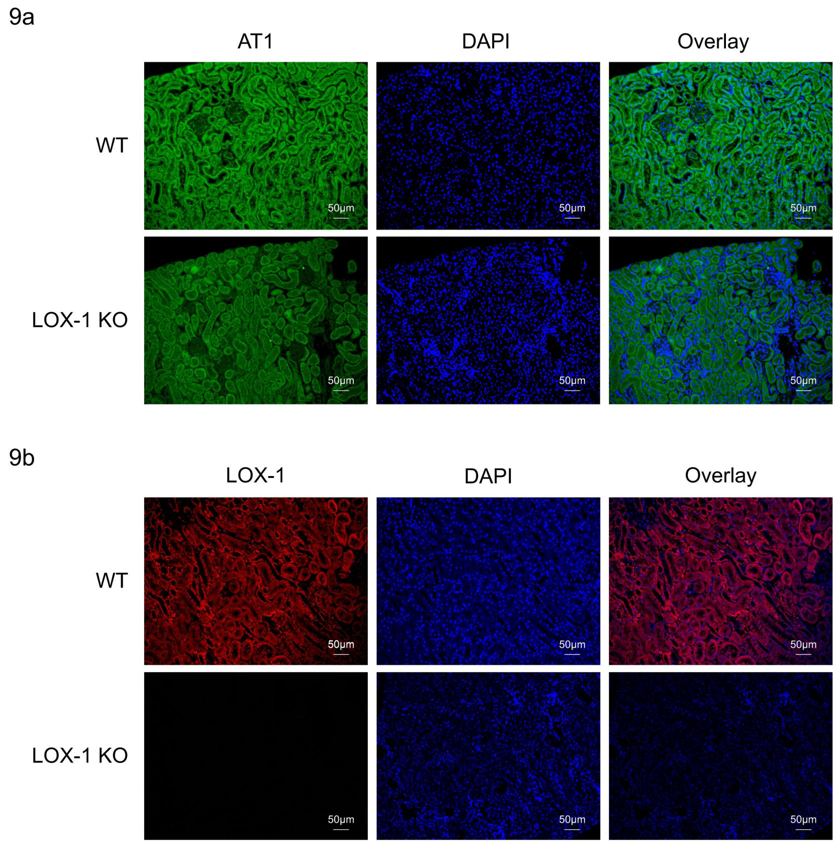


Fig. 9

LOX-1 and AT1a were predominantly co-localized in renal tubules

a, b Representative images depicting staining for LOX-1 (a) and AT1 (b) in the renal cortex tissues from wildtype mice (WT) and LOX-1 knockout mice (LOX-1 KO). Nuclei are stained blue with DAPI. Green and red signals indicates AT1, while the red signal indicates LOX-1. Overlay images demonstrate the merged visualization of AT1 or LOX-1 with DAPI, highlighting the predominant colocalization of LOX-1 and AT1 in renal tubules as opposed to the glomerulus.

oxLDL did not induce any additional influence on the internalization of both receptors upon AT1-β-arrestin activation compared to each ligand alone, suggesting that the quantity of activated receptors is not affected by the combination treatments. Considering the conformation-activation relationship in AT1 activation[20], this supports the hypothesis that the simultaneous binding of Ang II and oxLDL in a single AT1-LOX-1 complex induces a greater conformational change, resulting in a more open conformation of AT1 compared with each ligand alone. However, it is technically challenging to directly identify structural modifications of the receptor complex using techniques such as crystal structure analysis. Instead, we utilized AT1 conformational sensors capable of differentiating between the conformational changes induced by Ang II and biased AT1[20]. Interestingly, we observed that one of the two conformational sensors employed in this study detected an augmented response in the presence of the combination treatment compared with Ang II alone. Importantly, this enhanced response was significantly inhibited when an LOX-1 antibody was introduced, indicating the dependency of LOX-1 on this phenomenon. These results indicate that the combined treatment with Ang II and oxLDL in the presence of LOX-1 induces a unique conformational change in individual AT1 molecules, which differs from the conformational changes induced by each single treatment alone.

It is important to note that the ability of LOX-1 ligands to enhance Ang II-AT1 signaling is not commonly observed. This was corroborated by the observation that BSA-conjugated AGE, a recognized ligand of LOX-1[17], failed to augment the production of IP1 by Ang II compared to control BSA (**Fig. 1d**). While the exact reason for this discrepancy is not yet understood, it is noteworthy that the predicted particle size of oxLDL[25] is significantly larger at 250 Å compared to the maximum particle size of AGE-BSA[26], which is 120 Å. This suggests a potentially greater impact of oxLDL on the structural modifications within the AT1-LOX-1 complex. However, further structural analysis is required to validate this hypothesis.

Notably, the Gq bias resulting from combination treatment varied across the mammalian cells examined. Specifically, we observed a combinatorial effect exclusively in renal epithelial and fibroblasts, whereas vascular endothelial and smooth muscle cells did not display the same response. In these renal cells, we observed increased Gq signaling, along with other cellular phenomena such as calcium influx, changes in gene expression, and alterations in cellular characteristics, including EMT and cell proliferation. Specifically, EMT is an important cellular phenomenon in renal epithelial cells to acquire the phenotype of myofibroblasts and induce fibrosis by the excessive production and deposition of extracellular matrix (ECM) proteins, although there is a debate regarding the precise role of EMT in vivo[27]. Numerous studies have suggested that the EMT plays a critical role in Ang II-mediated renal fibrosis. EMT is also involved in the pathological transition of normal fibroblasts to myofibroblasts, which promotes fibrosis. We found that αSMA expression, a well-known indicator of EMT, was dramatically enhanced when renal epithelial cells and fibroblast cells were subjected to the combination treatment of Ang II and oxLDL, in comparison to individual treatments. Additionally, fibroblast proliferation, as assessed by the BrdU assay, was notably enhanced by the combined treatment, as opposed to each treatment administered separately. Interestingly, the use of 5 μg/mL oxLDL in our study showed a tendency to decrease proliferation, which was counteracted by the administration of an ARB but not a Gq inhibitor. These findings suggest that the presence of Ang II leads to a significant transformation in the function of oxLDL, primarily due to its altered influence within the AT1-LOX-1 complex. Taken together, these results suggest that the simultaneous binding of oxLDL to LOX-1 and Ang II to AT1 results in a Gq-biased shift in AT1 activation, leading to a cellular phenomenon that could potentially contribute to renal damage.

In an animal study, we introduced a biological environment in which both circulating Ang II and oxLDL (an LOX-1 ligand) were increased in mice. Previous studies using mice fed an HFD have consistently reported the onset of renal injury, as determined by various measurements[28–32]. In contrast, 6 weeks of HFD feeding without Ang II treatment did not alter renal function in our mice. This can be attributed to the lack of obesity induced by the high-fat diet in this study. We used this diet based on a previous study that confirmed increased circulating LOX-1 ligand levels without body weight gain in mice[33], and this diet decreased body weight in our experiments. Obesity is a well-established and clinically proven risk factor of renal dysfunction. The mechanisms underlying this association are complex and involve various factors other than lipid abnormalities, such as hemodynamic changes that affect kidney circulation and the impact of adipose tissue on the production of adipokines and other inflammatory mediators[34,35]. Consequently, we observed the influence of elevated lipid particle levels on renal function, independent of obesity. We found that the effect of an HFD became obvious with an increase in the Ang II load in WT mice. In particular, in WT mice treated with high-dose Ang II, which elevated systolic BP by approximately 30 mm Hg, an HFD induced notable increases in urinary reactive oxygen species and urinary albumin. In contrast, an HFD had no impact on Ang II-infused LOX-1 KO mice, as evidenced by equivalent urinalysis measurements for renal injury between the diets. Correspondingly, simultaneous administration of an HFD and Ang II resulted in a consistent alteration in the expression of genes related to renal injury, including fibrosis, inflammation, and oxidative stress, except for some genes in WT mice, but not in LOX-1 KO mice. This strongly suggests that the combined effect of Ang II and HFD on renal function is LOX-1 dependent. Nevertheless, it should be noted that the effect of high-dose Ang II infusion on BP tended to be less pronounced in LOX-1 KO mice compared to WT mice, although there were no differences in BP elevation between the diets in each group of mice. These findings align with those of previous studies indicating that LOX-1 knockout mice show resistance to Ang II-induced elevation of BP[24,36,37]. Specifically, when mice were infused with 2 γ Ang II (equivalent to 25 g mice) for a duration of 28 days, wildtype mice experienced a BP increase exceeding 180 mmHg, while LOX-1 knockout mice demonstrated a reduction of approximately 40 mmHg in this elevation[24]. Furthermore, the same research group reported that Ang II infusion led to less severe renal injury in LOX-1KO mice compared to WT mice[36]. Importantly, these findings were observed in mice fed with an ND, suggesting that the protective effect of LOX-1 loss-of-function against Ang II-induced elevated BP occurs through the LOX-1-AT1 complex, independent of the presence of oxLDL. Taken together, the effects of LOX-1 on AT1 signaling are complex, involving both ligand-dependent and -independent mechanisms, and further investigation is required for a comprehensive understanding of the LOX-1-AT1 interaction.

Finally, the current findings unequivocally demonstrated the molecular interactions between key molecules associated with dyslipidemia and hypertension in the kidneys. Moreover, this interaction can be effectively inhibited by ARBs, suggesting an additive effect in preventing the development of CKD, particularly in patients with hypertension and dyslipidemia. Interestingly, Ang II-dependent hypertensive animal models, including constriction of the renal artery and infusion of Ang II in rats and mice, have revealed a progressive increase in intrarenal Ang II levels, surpassing what can be accounted for by circulating Ang II levels alone[38]. This is due to Ang II-dependent renal activation of the renin-angiotensin system, as indicated by increased urinary angiotensinogen (AGT) secretion[38,39]. Importantly, the elevation of urinary AGT is also evident in patients with various pathologies, including hypertension and CKD, implying that renal Ang II levels increase even in individuals who do not exhibit elevated levels of circulating Ang II[21,39,40]. Taken together, the current finding of the synergistic effect of Ang II and oxLDL on AT1 activation in renal tissue is highly relevant for the development of kidney disease. RA system inhibitors, ARB, and ACE inhibitors are prioritized therapies to prevent the development of CKD with proteinuria in patients with hypertension[41]. In addition to the well-established inhibitory effects of Ang II on the contraction of efferent arterioles[42], a novel renal protective action of RA system inhibitors has been proposed. Particularly in hypertension accompanied by dyslipidemia, RA system inhibitors may exhibit anti-inflammatory, antifibrotic, and antioxidant effects in the kidneys by inhibiting the Gq signaling pathway through the AT1-LOX-1 complex in renal tubular cells and fibroblasts. Collectively, the current findings suggest that RA inhibitors can concomitantly reduce the effect of increased renal Ang II and oxLDL levels on the development of CKD in patients with hypertension and dyslipidemia, although direct evidence in clinical studies to support this remains to be elucidated.

This study has several limitations as follows: 1) The kidney, a complex organ vital for maintaining homeostasis, comprises a myriad of distinct cell types working in concert to execute its multifaceted functions[43]. The experiments conducted in mice raised questions regarding the specific cell types implicated in the synergistic effect of Ang II and oxLDL within the LOX-1-AT1 complex in the kidney. Addressing this issue would ideally require single-cell analysis, which is a challenge for future research. 2) The study employed systemic LOX-1 knockout mice. For a more detailed analysis, a phenotypic investigation using renal tissue-specific LOX-1 and AT1 knockout mice is required. Of particular importance is the analysis using tubule-specific knockout mice, in which the localization of these components has been verified through immunohistochemical staining. 3) The administration of Ang II (both pressor and subpressor doses) and its combination with HFD did not result in any histological changes in terms of fibrosis and mesangial expansion, despite the observed alterations in the associated gene expression and urinalysis for renal injury. Alterations in gene expression and urinalysis for renal injury may be sensitive and relatively early phenomena, and this discrepancy could potentially be attributed to the relatively short intervention duration of 4 weeks. Therefore, concurrent pathohistological alterations in the renal tissue might become evident with a more extended intervention period. 4) This study was limited by its inability to detect the amplification of Gq signaling in mouse renal tissue due to the concurrent administration of Ang II and HFD. Overcoming this limitation is a challenge for future studies.

In conclusion, the current findings suggest that the simultaneous binding of oxLDL and Ang II to their respective receptors within the complex induces a distinct conformational change compared with the effect of each ligand alone. This unique conformational change results in the heightened activation of G protein signaling and subsequent unfavorable cellular reactions in renal component cells. The relevance of this phenomenon was confirmed in mouse models, in which renal dysfunction was prominently exacerbated when there was a concomitant increase in Ang II and oxLDL levels. Notably, this effect was abolished by the deletion of LOX-1, indicating LOX-1 dependency of this in vivo phenomenon (**Fig. 10**). These findings indicate the clinical relevance of the direct interaction between hypertension and dyslipidemia and further support the clinical significance of RA inhibition in treating patients with CKD.

Materials and methods

Cell culture and materials

HUVECs and BAECs were cultured in EGM-2 (Lonza, Basel, Switherlad). Cells with fewer than five passages were used in the experiments. Transgenic CHO cells were maintained in an F-12 Nutrient Mixture with GlutamaxTM-I (Thermo Fisher Scientific, MA, USA), 10% fetal bovine serum (FBS; Gibco, USA), and appropriate selection reagents, as described below. CHO-K1 cells were maintained in F-12 Nutrient Mixture with GlutamaxTM-I and 10% FBS. HAVSMCs were cultured in Dulbecco's modified Eagle's medium/F12 (DMEM/F12) (Nacalai Tesque, Japan) supplemented with 1% penicillin-streptomycin (Fujifilm, Japan) and 10% FBS. A10 cells were grown in DMEM (Wako, Osaka, Japan) supplemented with 10% FBS and 1% penicillin-streptomycin. NRK52E and NRK49F cells (ECACC, UK) were cultured in DMEM (Wako, Japan) supplemented with 5% FBS and the appropriate selection reagents. Gene transcription in CHO cells was induced by adding 100 ng/mL doxycycline (Merck KGaA, Darmstadt, Germany). Cells were incubated at 37°C in 5% CO_2 and 95% air.

Construction of plasmid vectors

For stable transformants, pTRE2hyg vector (Clontech, USA) encoding mutated hAT1 with impaired ability to activate β-arrestin (pTRE2hyg-HA-FLAG-hAT1mg) were created using site direct mutagenesis as previously described[14]. For real-time imaging, LOX-1 tagged with V5-6×His at the

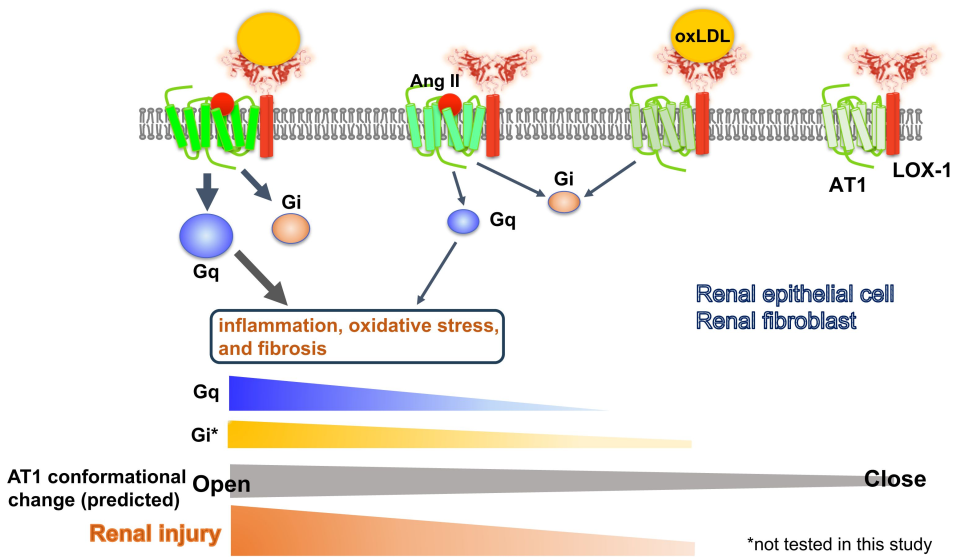


Fig. 10

Schematic overview of the AT1 and LOX-1 Interaction dynamics in renal cells

This schematic summary illustrates the predicted structure-activation relationship of the AT1 receptor within the LOX-1-AT1 complex in renal component cells. This highlights how the simultaneous binding of Ang II to AT1 and oxLDL to LOX-1 induces conformational changes in AT1. These changes were more pronounced than those triggered by the individual ligands. Such structural alterations have been proposed to amplify Gq signaling pathway activation, subsequently leading to renal damage.

C-terminus (V5-LOX-1) was subcloned into pmScarlet_C1 (plasmid #85042; Addgene) (mScarlet-LOX-1). HA-FLAG-hAT1 was subcloned into pcDNA3-EGFP (plasmid #85042; Addgene) (AT1-eGFP)[14].

Stable transformants

We constructed CHO-K1 cells expressing tetracycline-inducible human LOX-1 tagged with V5-6×His at the C-terminus (CHO-LOX-1), human HA-FLAG-hAT1 (CHO-AT1), or cells expressing both human LOX-1 and AT1 (CHO-LOX-1-AT1), as previously described[13,14]. To establish cells expressing both LOX-1 and mutated AT1 (pTRE2hyg-HA-FLAG-hAT1mg), they were co-transfected with the pSV2bsr vector (Funakoshi, Japan) into CHO-LOX-1 using the Lipofectamine 2000 transfection reagent (Thermo Fisher Scientific, USA). The stable transformants were selected with 400 μg/mL of hygromycin B (Wako, Osaka, Japan) and 10 μg/mL of blasticidin S (Funakoshi, Japan). The resistant clone expressing LOX-1 and mutated AT1 in response to doxycycline (Calbiochem, USA) was selected for use in the experiments (CHO-LOX-1-AT1mg) [14].

Small interfering RNA

NRK52 and NRK47 cells were plated at 50% confluence on the day of transfection. Silencer Select small interfering RNA (siRNAs) for *LOX-1* and *AT1a* (Thermo Fisher Scientific, MA, USA) were transfected into cells in medium without serum or antibiotics using Lipofectamine RNAiMAX (Thermo Fisher Scientific, MA, USA), according to the manufacturer's instructions.

Preparation of oxLDL

Human plasma LDL (1.019-1.063 g/mL), isolated by sequential ultracentrifugation, was oxidized using 20 μM $CuSO_4$ in PBS at 37 °C for 24 h. Oxidation was terminated by adding excess EDTA. LDL oxidation was analyzed by agarose gel electrophoresis for migration versus LDL[13].

Quantification of cellular IP1 accumulation

Gq-dependent activation of phospholipase C was quantified by measuring IP1 using the IP-One assay kit (Cisbio, France) as previously described[19]. Briefly, cells were seeded at 80,000 cells/well in 96 well transparent cell culture plates and incubated under serum-free conditions for 24 h. Thereafter, cells were treated for 1 h with IP1 stimulation buffer, including vehicle, native LDL, oxLDL, Ang II, AGE, PTX (Merck KGaA, Darmstadt, Germany), YM-254890 (Fujifilm Wako, Osaka, Japan), and RKI-1448 (Selleck, USA), as described in the text. Cell lysates with Triton X at a final concentration of 1% were transferred to a 384-well white plate, and IP1 levels were measured by incubating the cell lysates with FRET reagents (cryptate-labeled anti-IP1 antibody and d2-labeled IP1 analog).

Quantitative real-time PCR

RNA samples were purified using RNeasy Mini Kit (Qiagen, Germantown, MD, USA). One microgram of RNA was converted into cDNA using a ReverTra Ace qPCR RT kit (TOYOBO, Osaka, Japan) according to the manufacturer's instructions. All genes were evaluated using the ViiA7 Real-Time PCR System (Applied Biosystems, Thermo Fisher Scientific, Waltham, MA, USA). The data were analyzed using the ΔΔCt method with normalization against the GAPDH RNA expression in each sample. The primer sequences are listed in the Supplemental Table.

Western blotting

Proteins were separated using sodium dodecyl sulfate-polyacrylamide gel electrophoresis and electrophoretically transferred onto polyvinylidene fluoride membranes. The membranes were blocked with 5% nonfat dried milk and incubated with primary antibodies overnight at 4 °C. The primary antibodies used in this study were as follows: anti-SMA antibody (1:1000), anti-α-Tubulin antibody (1:1000) (Cell Signaling Technology, Inc, Danvers, MA, USA). The bands were visualized with a chemiluminescence detection system (LAS-4000 mini; GE Healthcare Life Sciences, Buckinghamshire, UK) using Chemi-Lumi One Super (Nacalai Tesque, Kyoto, Japan).

Calcium influx assay

Calcium influx was measured using Fura 2-AM (Dojindo, Kumamoto, Japan) with slight modifications to the manufacturer's protocol. In brief, cells plated in 96 wells were incubated with 5 μM Fura 2-AM in HEPES buffer saline (20 mM HEPES, 115 mM NaCl, 5.4 mM KCl, 0.8 mM MgCl2, 1.8 mM $CaCl_2$, 13.8 mM glucose, pH 7.4) for 1 h at 37°C, followed by replacement with recording medium without Fura 2-AM. Cells were treated with oxLDL, Ang II, or a combination of both at the indicated concentrations. Changes in F340/F380, an index of intracellular calcium concentration, were measured by dual-excitation microfluorometry using a digital image analyzer (Aquacosmos; Hamamatsu Photonics, Hamamatsu, Japan).

BrdU assay

Proliferative activity was assessed using a BrdU Cell Proliferation ELISA Kit (Funakoshi, Japan). NRK49F cells were seeded in 96-well tissue culture plates and incubated with the test reagents for 24 h. After incubating the cells with BrdU, we fixed the cells and denatured their DNA using a Fixing Solution. The plate was washed thrice with Wash Buffer before adding the Detector Antibody. Next, 100 μL/well of anti-BrdU monoclonal Detector Antibody was added, and the plate was incubated for 1 h at room temperature. Subsequently, 100 μL/well of Goat Anti-Mouse IgG Conjugate was pipetted and incubated for 30 minutes at room temperature. After five washes, the reaction was stopped by adding Stop Solution to each well. The color of the positive wells changed from blue to bright yellow. Finally, the plate was read at a wavelength of 450/550 nm using a spectrophotometric microtiter plate reader.

Creation of lentivirus encoding AT1 conformational sensors

For lentivirus encoding AT1 conformational sensors, rat AT1 and RlucII were subcloned into the pLVSIN-CMV Neo vector (Takara Bio, Japan). Next, the FlAsH binding sequence (CCPGCC) was inserted between residues K135 and S136 in the third intracellular loop (AT1-ICL3P3), as well as between residues K333 and M334 in the cytoplasmic-terminal tail (AT1-Ctail), utilizing the KOD-Plus Mutagenesis Kit (Toyobo, Japan), at the same site as previously documented[14].

Intramolecular FlAsH Bioluminescence resonance energy transfer assay to detect AT1 conformational change

CHO-LOX-1 cells were initially plated onto a white clear-bottom 96-well culture plate at a density of 1×10^5 cells/well. The following day, the cells were transduced with lenti-virus encoding AT1-Ctail or AT1-ICL3P3 [20] in 10% FBS. After 24 h of transduction, the cultures were transferred to serum-free conditions and incubated for an additional 24 h. Add 1.5 μM FlAsH-EDT_2 labeling reagent of TC-FlAsH™ II In-Cell Tetracysteine Tag Detection Kit (Thermo Fisher Scientific, MA, USA), wash twice with 250 μM BAL buffer, and assays were promptly conducted on a Spark® microplate reader (TECAN, Switzerland). The BRET ratio (emission mVenus/emission Rluc) was calculated as follows: Following a 3-minute baseline reading (with the final baseline reading presented at 0), cells were exposed to vehicle, oxLDL alone, AII alone, a combination of AII and oxLDL, or a combination of AngII, oxLDL, and LOX-1 antibodies. The BRET ratios were calculated every 16 seconds for a total of 320 seconds and the relative change in intramolecular BRET ratio was calculated by subtracting the average BRET ratio measured for cells stimulated with vehicle at each time point.

Analysis of LOX-1 and AT1 dynamics by real-time imaging

Live imaging was performed using previously reported methods[19]. Briefly, twenty-four hours prior to imaging experiments, CHO-K1 cells were transfected with LOX-1-mScarlet and AT1-eGFP by electroporation. Subsequently, the cells were seeded in a 35-mm glass base dish (Iwaki, Japan) that had been pre-coated with a 1000X diluted solution of 10 mg/mL poly-L-lysine (ScienCell, USA) one hour before seeding. The growth medium was substituted with imaging buffer (pH 7.4), which consisted of 125 mM NaCl, 5 mM KCl, 1.2 mM $MgCl_2$, 1.3 mM $CaCl_2$, 25 mM HEPES, and 3 mM D-glucose, with the pH adjusted to 7.4 using NaOH. Dynamic images of the cells were acquired at 25 °C using a SpinSR10 inverted spinning disk-type confocal super-resolution microscope (Olympus, Japan). The microscope was equipped with a 100x NA1.49 objective lens (UAPON100XOTIRF, Olympus, Japan) and an ORCA-Flash 4.0 V2 scientific CMOS camera (Hamamatsu Photonics KK, Japan) at 5-second intervals. The imaging experiment was conducted using CellSens Dimension 1.11 software, employing a 3D deconvolution algorithm (Olympus, Japan), and the number of puncta was determined using ImageJ1.53K[19].

Animals and diets

Male WT mice (C57BL/6J) and LOX-1 KO mice with a C57BL/6 background were used in this study. LOX-1 KO mice were generated as described previously[44]. Mice were housed in a temperature-controlled (20-22°C) room on a 12 hours light/dark cycle and fed an ND (MF; Oriental Yeast, Osaka, Japan) or an HFD (High Fat Diet without DL-α-tocopherol, CLEA Japan Inc, Tokyo, Japan), which reported to increase plasma LOX-1 ligand in ApoE KO mice[21]. All study protocols were approved by the Animal Care and Use Committee of Osaka University and were conducted according to the guidelines of the NIH for the Care and Use of Laboratory Animals.

Blood pressure measurement in mice

The blood pressure of the mice was measured using the tail-cuff method with BP-98A (Softron, Japan). The measurements were performed after restraining the mice. The blood pressure was calculated as the average of 6 readings for each animal at each time point.

Urine tests in mice

Urine tests in mice included the measurement of urine 8-OHdG, creatinine, and albumin concentrations. The DNA Damage (8-OHdG) ELISA Kit (StressMarq Bioscience, Canada), Creatinine Kit L type Wako (Fujifilm, Japan), and Mouse Albumin ELISA Kit (Bethyl Laboratories, Inc., TX, USA) were utilized for these measurements, following their respective instructions.

Plasma LOX-1 ligand concentration

Measurement of LOX-1 ligands containing apoB (LAB) in mouse plasma was performed using a modified protocol based on a previously reported method [33]. Briefly, recombinant human LOX-1 (0.25 μg/well) was immobilized on 384 well plate (greiner, Frickenhausen, Germany) by incubating overnight at 4°C in 50 μl of PBS. After three washes with PBS, 80 μl of 20% (v/v) ImmunoBlock (KAC, Kyoto, Japan) was added, and the plates were incubated for 2 h at 25°C. After three washes with PBS, the plates were incubated for 2 h at 25°C with 40 μl of standard oxidized LDL or samples. Samples were prepared by 4-fold dilution of plasma with HEPES-EDTA buffer (10 mM HEPES, 150 mM NaCl, 2 mM EDTA, pH 7.4), and standards were prepared by dilution of oxidized LDL with HEPES-EDTA buffer. Following three washes with PBS, the plates were incubated for 1 h at 25°C with chicken monoclonal anti-apoB antibody (HUC20, 0.5 μg/mL) in HEPES-EDTA containing 1% (w/v) BSA. After three washes with PBS, the plates were incubated for 1 h at 25°C with peroxidase-conjugated donkey anti-chicken IgY (Merck, NJ, USA) diluted 5000 times with HEPES-EDTA containing 1% (w/v) BSA. After five washes with PBS, the substrate solution containing 3,3',5,5'-tetramethylbenzidine (TMB solution, Bio-Rad Laboratories, CA, USA) was added to the plates and incubated them for 30 min at room temperature. The reaction was terminated with 2 M sulfuric acid. Peroxidase activity was determined by measuring absorbance at 450 nm using a SpectraMax 340PC384 Microplate Reader (Molecular Devices, CA, USA).

Tissue preperation

Kidneys were perfused with cold PBS before removal. Kidney samples were rapidly excised. A quarter of samples were stored at 4°C in RNAlater (Thermo Fisher Scientific, MA, USA) for RNA extraction. The remaining quarters were fixed in 4% paraformaldehyde overnight at 4°C for histological evaluation. The remaining half was snap-frozen in liquid nitrogen and stored at −80°C for further analysis.

Periodic acid-Schiff and Masson-Trichrome staining

The degree of glomerular mesangial expansion and glomerular area (representing the structural integrity of the glomeruli) were assessed in a blinded manner using periodic acid-Schiff (PAS) staining. Collagen accumulation was determined by Masson-Trichrome (MTC) staining. For MTC staining, the area displaying fibrosis was quantitatively evaluated in a blinded manner by measuring the blue staining in six strongly magnified fields of view using the ImageJ software, and the average was calculated after determining the ratio of the total area.

Fluorescent immunostaining

For fluorescent immunostaining of LOX-1 and AT-1, the mice were perfused with cold saline before tissue removal. After 3 days of zinc fixation, the tissue was replaced with 70% ethanol. The 3-μm-thick kidney tissue sections were immunohistochemically stained with antibodies against ATGR (1:200, Cosmo Bio, Japan) and OLR-1 (1:200, TS58 from the laboratory of T.S., Shinshu University School of Medicine, Nagano, Japan). Following deparaffinization (using Lemosol and gradient ethanol) and rehydration, the slices were subjected to antigen retrieval by autoclaving in citrate buffer (0.01 M; pH 6.0). Subsequently, the slices were washed thrice with PBS and blocked with 5% bovine serum albumin for 30 min at room temperature. The slides were then incubated with primary antibodies for 2 h at room temperature. Goat anti-Rabbit IgG (H+L) High Cross-Adsorbed Secondary Antibody, Alexa Fluor Plus 488 and 594 (Thermo Fisher Scientific, MA, USA) were used as secondary antibodies for ATGR and OLR-1, respectively. After incubation with secondary antibodies for 1 h at room temperature, the slices were washed with PBS. Finally, slides were sealed and photographed. Visual analyses were performed using a BZ-800L microscope (Keyence, Japan).

Statistical analyses

All data are presented as the mean ± SEM. Differences between two treatments or among multiple treatments were determined using the Student’s t-test or one-way ANOVA followed by Tukey’s multiple comparison test.

Data availability

All data supporting the findings of this study are available within the paper and its Supplementary Information. Source data are provided in this paper.

Acknowledgements

This work was partially supported by JSPS KAKENHI Grant Numbers 21K07389 (Y.T.), 22K08181 (Y.N.), 20H03576 (H.R.), and 18H02732 (K.Y.). We are grateful to Tomoko Sato, Yoshinori Koishi, and Chika Takana for technical assistance. We would like to thank Editage (*www.editage.com*) for the English language editing.

Author contributions

J.I., Y.H., Y.T., Y.N., T.T., and K.Y. designed the study. J.I. and Y.H. performed most of the experiments. A.K. measured plasma LOX-1 ligand concentrations in mice and assisted with data analysis. J.I. and S.S. performed histological analysis of the renal tissue of mice. J.I., Y.H., Y.T., Y.N., T.T., and K.Y. wrote the manuscript and analyzed the data. C.W., Z.W., G.Y., L.W., Y.N., R.O., T.F., H.K., K.A., H.T., S.S., and K.I. analyzed the data and contributed to the discussion. A.K., Y.I., H.R., and T.S. contributed to the discussion and reviewed and edited the manuscript. All authors contributed to the editing of the final manuscript. Y.T. and Y.N. had full access to all data in the study and took responsibility for the integrity of the data and accuracy of the data analysis.

Competing interests

The authors declare no competing interests.

Supplemental Figure Legends

Fig. S1

Live-imaging analysis of membrane LOX-1 and AT1 in response to the co-treatment of oxLDL with AngII.

Real-time membrane imaging of CHO cells co-transfected with LOX-1-mScarlet and AT1-eGFP in response to oxLDL (10μg/ml) in the combination of Ang II (10^{-7}M) (Supplemental Video). A count of puncta was performed using separate images visualizing LOX-1-mScarlet (red puncta) and AT1-eGFP (green puncta) immediately before and 3 min after ligand application.

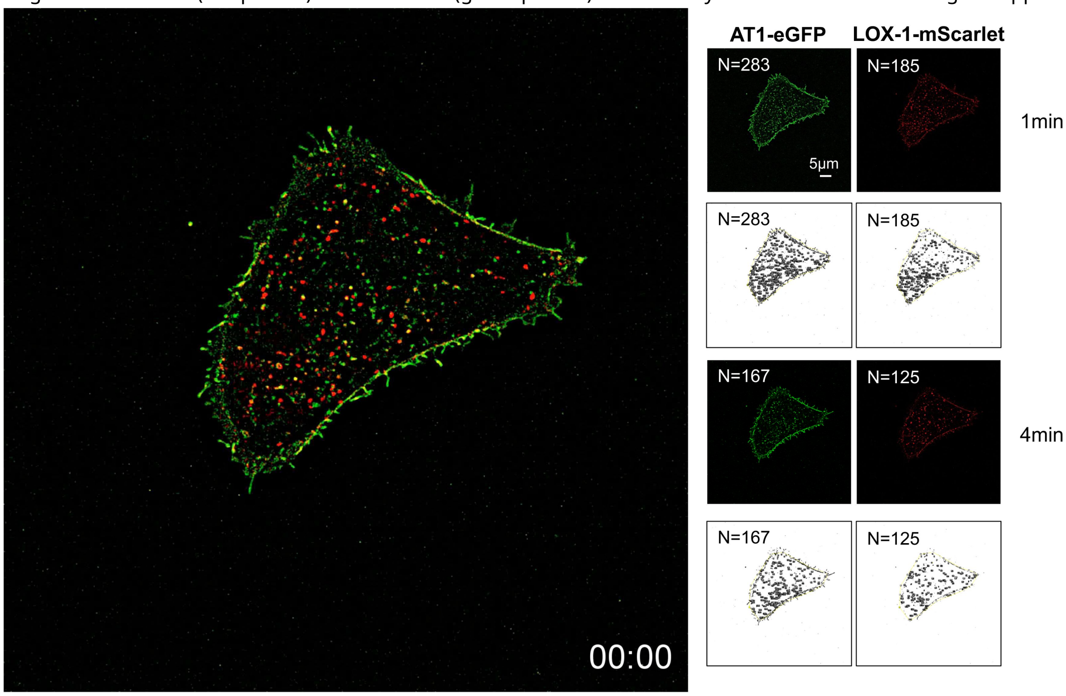


Fig. S2

Oxidized LDL in combination with Ang II do not increase cellular IP1 content in human umbilical vein endothelial cells and bovine vascular endothelial cells, human aortic vascular smooth muscle cells, and rat macrophages

IP1 concentration in response to oxLDL (10μg/ml) in the combination of Ang II (10^{-7}M) in HUVECs (human umbilical vein endothelial cell), BAECs (bovine aortic endothelial cell), HAVSMCs (human aortic vascular smooth muscle cell), and A10 cells (rat macrophages).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test (n=5 for each group).

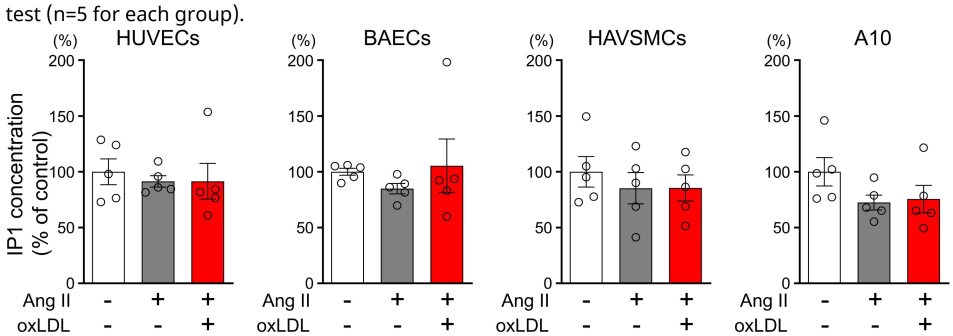


Fig. S3

Efficiency of siRNA-mediated knockdown for *AT1a* and *LOX-1* in NRK52E and NRK49F cells

NRK52E and NRK49F cells were transfected with siRNAs against scrambled siRNA, *AT1a* or *LOX-1*. The efficiency of siRNA-mediated gene silencing was quantified by assessing *AT1a* and *LOX-1* expression levels using quantitative real-time PCR. Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.

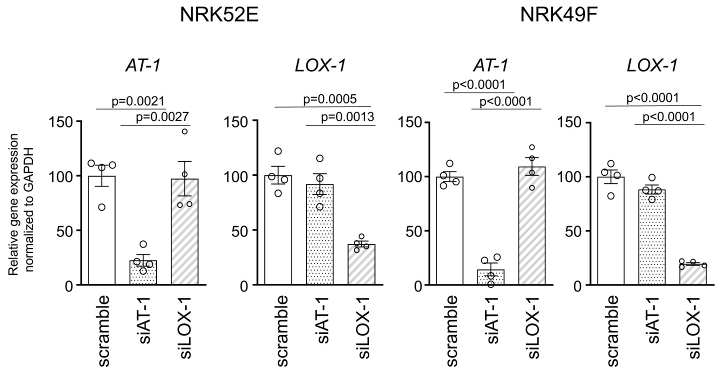
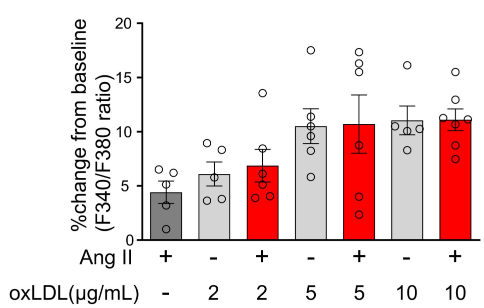


Fig. S4

Calcium influx was not induced by either the combination treatment of Ang II or oxLDL or each treatment alone in NRK52E cells

Percentage changes from baseline in the ratio of emission signals (F340/F380) measured by Fura 2-AM were quantified following treatment with Ang II (10^{-7}M) and oxLDL at the concentrations detailed in the figure for NRK49E cells (n=5-7 for each group).

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.



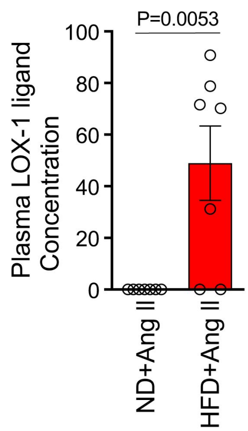


Fig. S5

High Fat Diet used in the study prominently increased plasma LOX-1 ligand concentration

Plasma LOX-1 ligand concentration of 14-week-old wildtype mice upon ND and HFD for 6 weeks. From 10 weeks of age, these mice also received concurrent 4-week infusions of Ang II at a pressor dose of 0.7 γ, administered through subcutaneously implanted osmotic pumps.

Data are represented as mean ± SEM. Differences were determined using Student's t-test (n=7 for each group).

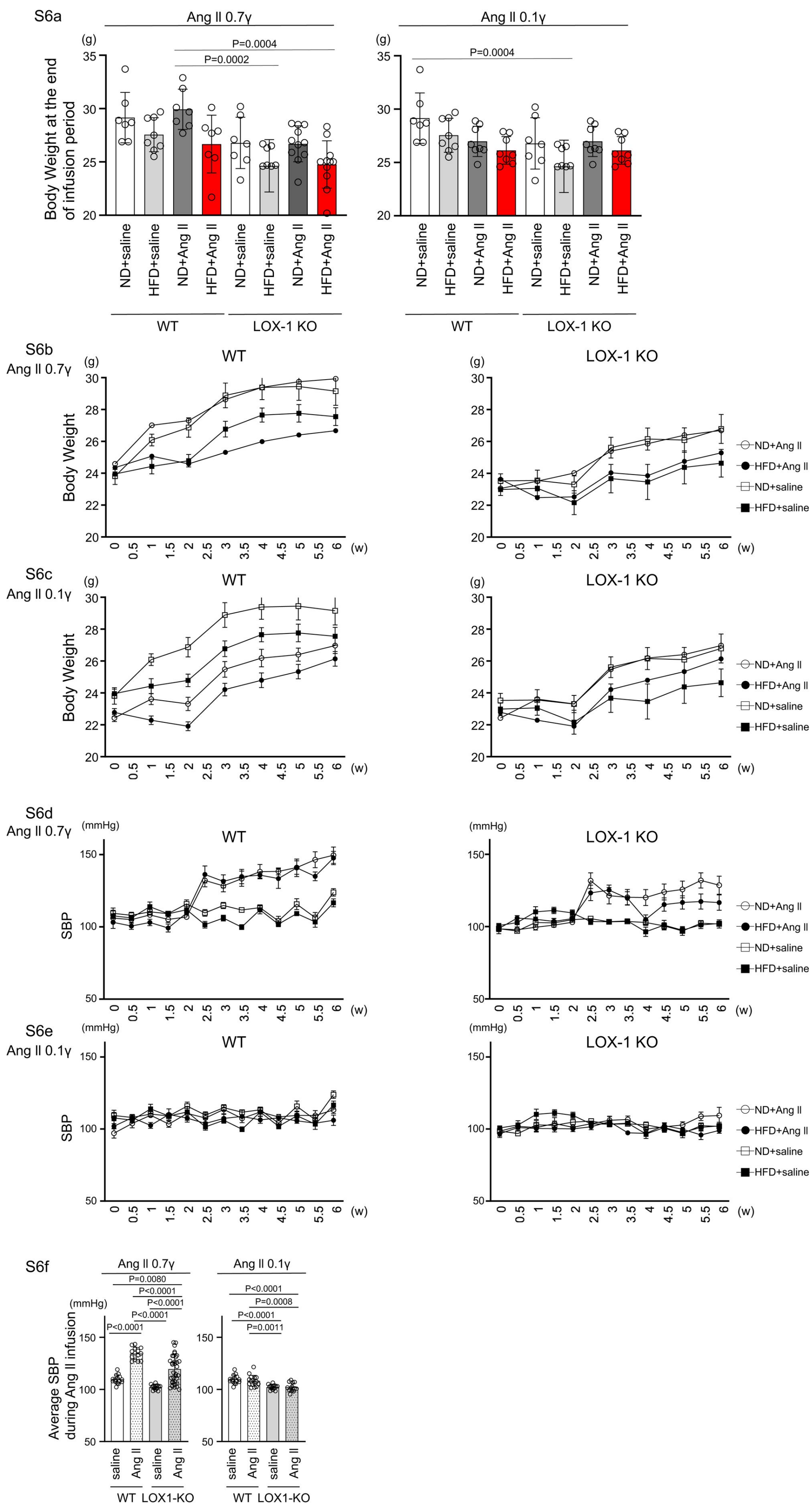

S6a
Ang II 0.7γ
Ang II 0.1γ
P=0.0004
P=0.0002
P=0.0004
Body Weight at the end of infusion period
(g)
ND+saline
HFD+saline
ND+Ang II
HFD+Ang II
WT
LOX-1 KO
S6b
Ang II 0.7γ
S6c
Ang II 0.1γ
Body Weight
WT
LOX-1 KO
ND+Ang II
HFD+Ang II
ND+saline
HFD+saline
(w)
S6d
Ang II 0.7γ
S6e
Ang II 0.1γ
(mmHg)
SBP
S6f
Ang II 0.7γ
Ang II 0.1γ
P=0.0080
P<0.0001
P<0.0001
P<0.0001
P<0.0001
P<0.0001
P=0.0008
P<0.0001
P=0.0011
Average SBP during Ang II infusion
saline
Ang II
WT
LOX1-KO

Fig. S6

Impact of Diet and Ang II Infusion on Body Weight and Systolic Blood Pressure in Mice

a Final Body weight: This figure shows the body weights of WT and LOX-1 KO mice at the end of the 4-week infusion period, highlighting the effects of diet and pharmacological treatment.

b, c Serial Body Weight Changes: These graphs depict the progression of body weight over time in WT and LOX-1 KO mice, illustrating the impact of the dietary regimen and Ang II infusion on weight dynamics.

d, e Systolic Blood Pressure Trajectory: Serial measurements of systolic blood pressure (SBP) obtained using the tail-cuff method are presented for WT and LOX-1 KO mice.

The figures show the changes in SBP over the course of the study, corresponding to the administration of a pressor dose of 0.7 γ (d) and a subpressor dose of 0.1 γ (e) of Ang II, respectively.

f Final SBP: This figure shows SBP of WT and LOX-1 KO mice at the end of the 4-week infusion period, highlighting the effects of diet and pharmacological treatments.

Beginning at 8 weeks of age, mice were fed either an ND or an HFD for 6 weeks. From 10 weeks of age, coinciding with the 2-week time point in the figure, the rats underwent a 4-week period of infusion with either vehicle or Ang II. The infusion was delivered at specific dosage levels through subcutaneously implanted osmotic pumps.

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).

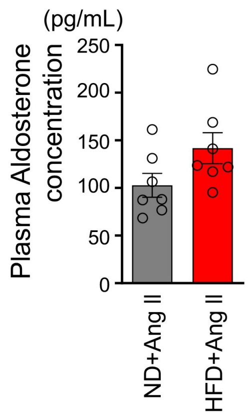


Fig. S7

No significant difference was found in plasma aldosterone concentration between a normal diet and a high fat diet-fed wildtype mice with a pressor dose of Ang II

Plasma aldosterone concentration in 14-week-old wild-type mice fed an ND or an HFD for 6 weeks. From 10 weeks of age, these mice also received concurrent 4-week infusions of Ang II at a pressor dose of 0.7 γ, administered through subcutaneously implanted osmotic pumps.
Data are represented as mean ± SEM. Differences were determined using Student's t-test (n=7 for each group).

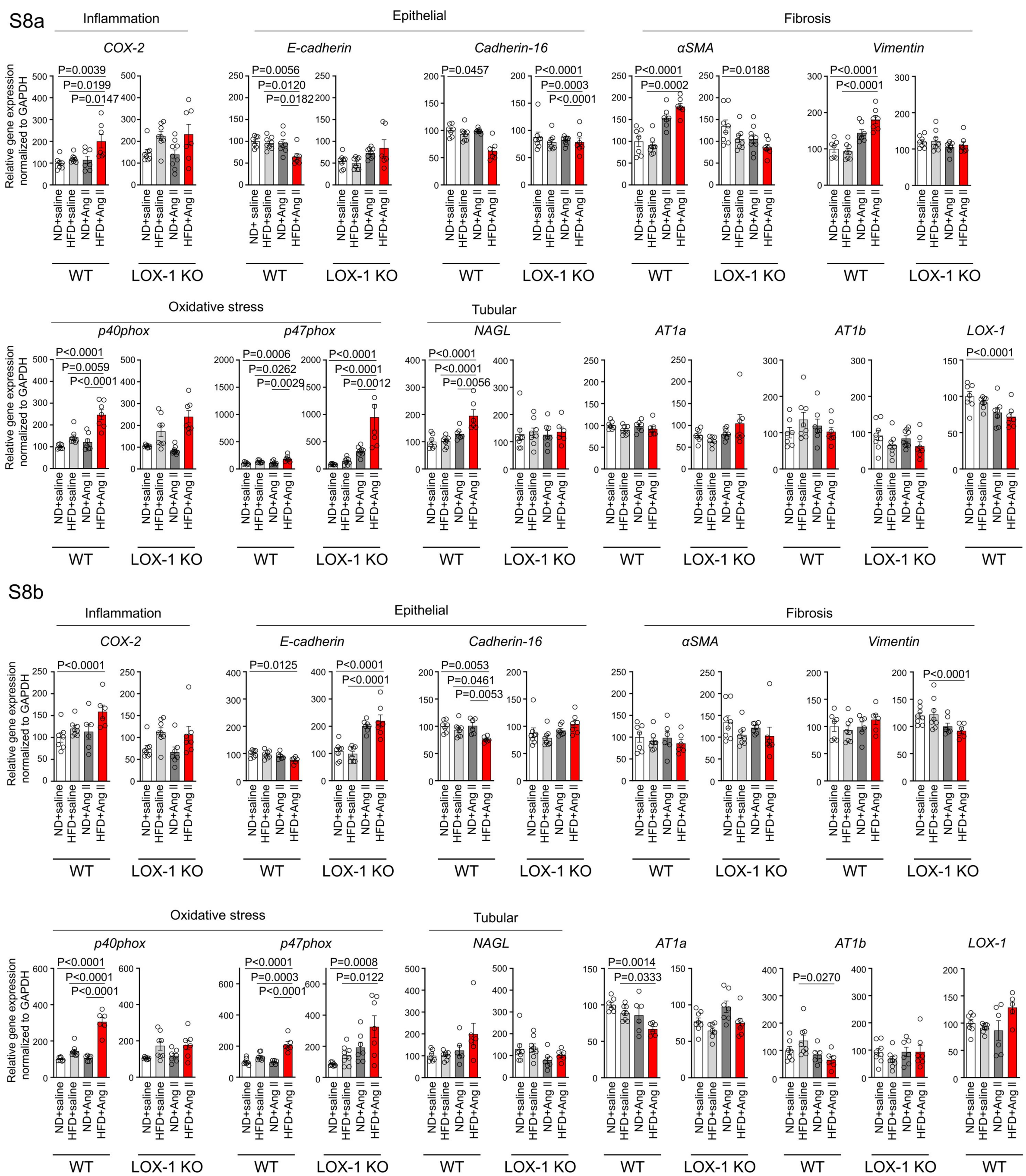


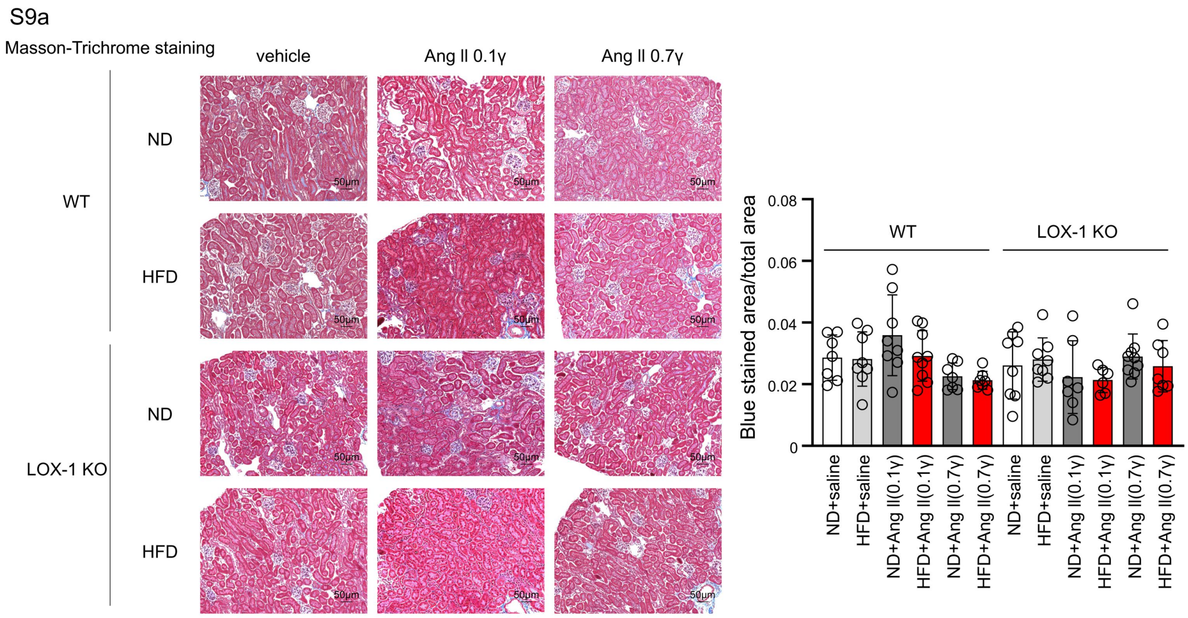
Fig. S8

A high-fat diet enhanced Ang II-induced renal injury-related gene expression in the kidney in a LOX-1-dependent manner.

Quantitative real-time PCR analysis for gene expression of NADPH components (*p40phox and p47phox*), inflammatory gene (*COX-2*), fibrosis markers (*αSMA and vimentin*), epithelial markers (*E-cadherin and cadherin-16*), tubular marker (*NAGL*), *AT1a*, *AT1b*, and *LOX-1* in the kidney harvested from WT and LOX-1 KO mice.

The experimental procedures, including the dietary regimen and Ang II administration, are detailed in **Fig. 7a**.

Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test for (a) and (b).



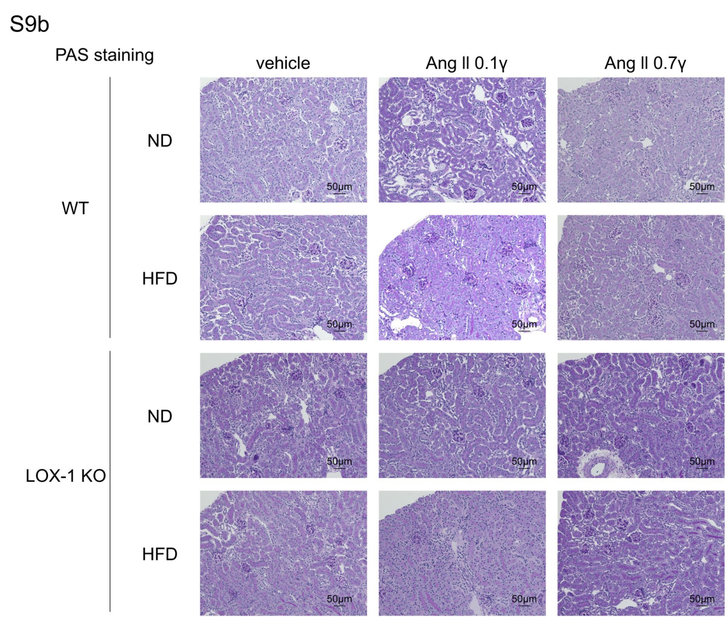


Fig. S9

The treatment with Ang II, a High Fat Diet, or their combination for 4 weeks did not induce any histological changes indicative of renal injury

a Left: Representative histological images of Masson-Trichrome staining used to detect fibrosis in renal tissues harvested from WT and LOX-1 KO mice. Right: Quantitative analysis by Masson-Trichrome staining.
Data are represented as mean ± SEM. Differences were determined using one-way ANOVA, followed by Tukey's multiple comparison test.
b Representative histological images of renal tissues harvested from WT and LOX-1 KO mice stained for PAS to assess mesangial expansion and glomerular area, providing insight into the structural integrity of the glomeruli.
Eight-week-old mice were fed an ND or an HFD for 6 weeks. From 10 weeks of age, these mice were concurrently treated for 4 weeks with infusions of vehicle or Ang II (a subpressor dose of 0.1 γ or a pressor dose of 0.7 γ) through subcutaneously implanted osmotic pomps.

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Editors

Reviewing Editor
Arduino Mangoni
Flinders Medical Centre and Flinders University, Adelaide, Australia

Senior Editor
Matthias Barton
University of Zurich, Zurich, Switzerland

Reviewer #1 (Public Review):

Summary:

This study demonstrates a key role of oxLDL in enhancing Ang II-induced Gq signaling by promoting the AT1/LOX1 receptor complex formation. Importantly, Gq-mediated calcium influx was only observed in LOX1 and AT1 both expressing cells, and AT1-LOX1 interaction aggravated renal damage and dysfunction under the condition of a high-fat diet with Ang II infusion, so this study indicated a new therapeutic potential of AT1-LOX1 receptor complex in CKD patients with dyslipidemia and hypertension.

Strengths:

This study is very exciting and the work is also very detailed, especially regarding the mechanism of LOX1-AT1 receptor interaction and its impact on oxidative stress, fibrosis, and inflammation.

Weaknesses:

The direct evidence for the interaction between AT1 and LOX1 receptors in cell membrane localization is relatively weak. Here I raise some questions that may further improve the study.

Major points:

(1) The authors hypothesized that in the interaction of AT1/LOX1 receptor complex in response to ox-LDL and AngII, there should be strong evidence of fluorescence detection of colocalization for these two membrane receptors, both in vivo and in vitro. Although the video evidence for AT1 internalization upon complex activation is shown in Figure S1, the more important evidence should be membrane interaction and enhanced signal of intracellular calcium influx.

(2) Co-IP experiment should be provided to prove the AT1/LOX1 receptor interaction in response to ox-LDL and AngII in AT1 and LOX1 both expressing cells but not in AT1 only expressing cells.

(3) The authors mentioned that the Gq signaling-mediated calcium influx may change gene expression and cellular characteristics, including EMT and cell proliferation. They also provided evidence that oxidative stress, fibrosis, and inflammation were all enhanced after activating both receptors and inhibiting Gq was effective in reversing these changes. However, single stimulation with ox-LDL or AngII also has strong effects on ROS production, inflammation, and cell EMT, which has been extensively proved by previous studies. So, how to distinguish the biased effect of LOX1 or AT1r alone or the enhanced effect of receptor conformational changes mediated by their receptor interaction? Is there any better evidence to elucidate this point?

(4) How does the interaction between AT1 and LOX1 affect the RAS system and blood pressure? What about the serum levels of rennin, angiotensin, and aldosterone in ND-fed or HFD-fed mice?

https://doi.org/10.7554/eLife.98766.1.sa1

Reviewer #2 (Public Review):

Individuals with chronic kidney disease often have dyslipidemia, with the latter both a risk factor for atherosclerotic heart disease and a contributor to progressive kidney disease. Prior studies suggest that oxidized LDL (oxLDL) may cause renal injury through the activation of the LOX1 receptor. The authors had previously reported that LOX1 and AT1 interact to form a complex at the cell surface. In this study, the authors hypothesize that oxLDL, in the setting of angiotensin II, is responsible for driving renal injury by inducing a more pronounced conformational change of the AT1 receptor which results in enhanced Gq signaling.

They go about testing the hypothesis in a set of three studies. In the first set, they engineered CHO cell lines to express AT1R alone, LOX1 in combination with AT1R, or LOX1 with an inactive form of AT1R and indirectly evaluated Gq activity using IP1 and calcium activity as read-outs. They assessed activity after treatment with AngII, oxLDL, or both in combination and found that treatment with both agents resulted in the greatest level of activity, which could be effectively blocked by a Gq inhibitor but not a Gi inhibitor nor a downstream Rho kinase inhibitor targeting G12/13 signaling. These results support their hypothesis, though variability in the level of activation was dramatically inconsistent from experiment to experiment, differing by as much as 20-fold. In contrast, within the experiment, differences between the AngII and AngII/oxLDL treatments, while nominally significant and consistent with their hypothesis, generally were only 10-20%. Another example of unexplained variability can be found in Figures 1g-1j. AngII, at a concentration of 10-12, has no effect on calcium flux in one set of studies (Figure 1g, h) yet has induced calcium activity to a level as great as AngII + oxLDL in another (Figure 1i). The inconsistency of results lessens confidence in the significance of these findings. In other studies with the LOX1-CHO line, they tested for conformational change by transducing AT1 biosensors previously shown to respond to AngII and found that one of them in fact showed enhanced BRET in the setting of oxLDL and AngII compared to AngII alone, which was blocked by an antibody to AT1R. The result is supportive of their conclusions. Limiting enthusiasm for these results is the fact that there isn't a good explanation as to why only 1 sensor showed a difference, and the study should have included a non-specific antibody to control for non-specific effects.

The authors then repeated similar studies using publicly available rat kidney epithelial and fibroblast cell lines that have an endogenous expression of AT1R and LOX1. In these studies, oxLDL in combination with AngiI also enhanced Gq signaling, while knocking down either AT1R or LOX1, and treatment with inhibitors of Gq and AT1R blocked the effects. Like the prior set of studies, however, the effects are very modest and there was significant inter-experimental variability, reducing confidence in the significance of the findings. The authors then tested for evidence that the enhanced Gq signaling could result in renal injury by comparing qPCR results for target genes. While the results show some changes, their significance is difficult to assess. A more global assessment of gene expression patterns would have been more appropriate. In parallel with the transcriptional studies, they tested for evidence of epithelial-mesenchymal transition (EMT) using a single protein marker (alpha-smooth muscle actin) and found that its expression increased significantly in cells treated with oxLDL and AngII, which was blocked by inhibition of Gq inhibition and AT1R. While the data are sound, their significance is also unclear since EMT is a highly controversial cell culture phenomenon. Compelling in vivo studies have shown that most if not all fibroblasts in the kidney are derived from interstitial cells and not a product of EMT. In the last set of studies using these cell lines, the authors examined the effects of AngII and oxLDL on cell proliferation as assayed using BrdU. These results are puzzling---while the two agents together enhanced proliferation which was effectively blocked by an inhibitor to either AT1R or Gq, silencing of LOX1 had no effect.

The final set of studies looked to test the hypothesis in mice by treating WT and Lox1-KO mice with different doses of AngII and either a normal or high-fat diet (to induce oxLDL formation). The authors found that the combination of high dose AngII and a high-fat diet (HFD) increased markers of renal injury (urinary 8-ohdg and urine albumin) in normal mice compared to mice treated with just AngII or HFD alone, which was blunted in Lox1-KO mice). These results are consistent with their hypothesis. However, there are other aspects of these studies that are either inconsistent or complicating factors that limit the strength of the conclusions. For example, Lox1- KO had no effect on renal injury marker expression in mice treated with low-dose AngII and HFD. It also should be noted that Lox1-KO mice had a lower BP response to AngII, which could have reduced renal injury independent of any effects mediated by the AT1R/LOX1 interaction. Another confounding factor was the significant effect the HFD diet had on body weight. While the groups did not differ based on AngII treatment status, the HFD consistently was associated with lower total body weight, which is unexplained. Next, the authors sought to find more direct evidence of renal injury using qPCR of candidate genes and renal histology. The transcriptional results are difficult to interpret; moreover, there were no significant histologic differences between groups. They conclude the study by showing the pattern of expression of LOX1 and AT1R in the kidney by immunofluorescence and conclude that the proteins overlap in renal tubules and are absent from the glomerulus. Unfortunately, they did not co-stain with any other markers to identify the specific cell types. However, these results are inconsistent with other studies that show AT1R is highly expressed in mesangial cells, renal interstitial cells, near the vascular pole, JG cells, and proximal tubules but generally absent from most other renal tubule segments.

In sum, this study tackles an important clinical issue and provides some in vitro evidence to support a mechanism whereby dyslipidemia could accelerate renal functional decline through activation of the AT1R/LOX1 complex by oxLDL and AngII.

However, a very high degree of variability in the results, modest within-experiment differences, some internal inconsistencies that aren't explained, and the lack of compelling and strongly supportive in vivo results suggest this is still more a hypothesis than an established likely mechanism.

https://doi.org/10.7554/eLife.98766.1.sa0